



Mathematical Modelling of Malaria Integrating Temperature, Rainfall, and Vegetation Index

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Abstract

Environmental factors such as temperature, rainfall, and vegetation index play a crucial role in the transmission dynamics of malaria. Accurately quantifying the complex relationships between these variables and the malaria burden poses a significant challenge. In this study, we developed a host-mosquito mathematical model to investigate the impact of temperature, rainfall, and normalized difference vegetation index (NDVI) on malaria transmission dynamics calibrated with the Burundi case's study. Mathematical analysis explored the equilibria, stability, and computation of the model's threshold values. Numerical simulations suggest that temperature, rainfall, and vegetation index affect the transmission dynamics of malaria. Temperature and NDVI appear to exhibit a more pronounced influence among these factors. The conditions conducive to malaria transmission include a mean monthly temperature range of [20–25 °C], an averaged monthly NDVI range of approximately [0.4–0.6]. The reproduction number was used as a quantitative measure to predict the impact of temperature, rainfall, and NDVI on malaria transmission dynamics across Burundi. The results suggest a progressive increase in the reproduction number over time, suggesting a threat of the rising number of cases in Burundi if drastic control measures are not implemented.

Keywords Mathematical model · Malaria · Reproduction number · Temperature · Rainfall · NDVI

1 Introduction

The effects of environmental factors on human health have raised significant concerns. The direct effects of rapid fluctuations in environmental parameters such as temperature, rainfall,

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and vegetation index on malaria transmission have been of particular interest as they challenge current control and eradication strategies. Malaria is one of the major public health problems globally. It is an infectious disease caused by plasmodium protozoan parasites that is transmitted between humans through the bite of the female anopheles mosquito. Female anopheles mosquitoes transmit malaria by collecting plasmodium parasites when they bite an infected person. The parasites develop and multiply within the mosquito and migrate to its salivary glands. Throughout this study, the term mosquito is used to refer to the female anopheles mosquito. When the mosquito bites a new person, the parasites are injected into their bloodstream, causing a malaria infection [1–3]. Millions of people are infected by malaria every year. Sub-Saharan Africa continues to bear the heaviest burden of malaria globally, accounting for 95% of all malaria cases in the world [4]. Recent malaria statistics reveal that approximately 223 million cases were recorded in 2022, representing 94% of the global burden. The region accounted for 95% of all malaria deaths globally in 2022, with about 78% of these deaths occurring among children under the age of five [5].

Burundi, an East African country experienced an increase in the malaria epidemic in recent years and notably from 2.6 million in 2013 to 8.3 million in 2016 [6]. With an approximate population of 12,890,000, it is estimated that 80% of the Burundi's population is susceptible to malaria infection [7]. Approximately 56% of the population live in potentially epidemic areas, while 23% live in hyper-endemic areas [8]. Each year, around 2 million cases of malaria on average are identified throughout the country. Pregnant women and children below the age of 5 represent the demographic most afflicted category. In these age groups, 48% of deaths in health facilities are attributable to malaria [8]. On record, about 50% of the outpatient consultations are due to malaria [9]. Initial assessments suggest that several primary factors contribute to the current increase in malaria transmission. These factors include reduced utilization of preventive measures, diminished population immunity, and lower socioeconomic status. Additionally, heightened mobility, especially among residents of mountainous regions, plays a significant role. Usually, malaria incidence remains low in these regions, but changes in climate patterns and alterations in vector ecology and behavior are causing shifts in the disease's prevalence [8].

In the past two decades, analysis of the universal distribution of the disease shows that it is more concentrated in warm regions. However, due to climate change and global warming, formerly sheltered areas are now being invaded. Climate plays a vital role in the increase in malaria transmission in Africa because temperature and rainfall levels are predominantly high compared to other continents [10]. The increment in the number of mosquitoes depends on the climatic conditions and the accessibility of oviposition sites, which in turn are subordinate to natural and environmental factors [11]. Several authors have shown that environmental and climatic factors significantly influence the dynamic transmission of malaria [12–14]. During high temperatures, the mosquitoes' digestive systems become agile which increases the number of bites and the increase of the disease [15]. Variations in temperature, humidity, rainfall, and wind speed impact the occurrence of malaria. These changes can either affect the lifespan of mosquitoes or influence the behaviors of humans, vectors (mosquitoes), or parasites, thereby influencing the transmission of the disease [16, 17]. The mosquito population is very sensitive to climatic conditions; a small temperature change can affect the life span of mosquitoes. Mosquitoes, as ectothermic organisms, have a body temperature that is dependent on the external environment. This relationship between their body temperature and the environment significantly impacts their metabolic rate. Warmer temperatures tend to speed up their metabolism, leading to quicker development and shorter lifespans. In contrast, cooler temperatures have the opposite effect, slowing down their metabolism and extending their life cycle [18, 19]. This forms the basis for its perpetual

prevalence in the tropical climate because there are enough breeding sites and conducive temperatures.

Vegetation is a climatic environmental factor that influences the behavior of mosquito vectors both directly and indirectly [20]. The vegetation index is used to assess the density of vegetation in a particular area and the most frequently used as vegetation indices is the NDVI (Normalized Difference vegetation indices). It is calculated as the normalized difference between the electromagnetics spectrum's near Infrared (NIR) and red bands. The NDVI scale runs from -1 to $+1$ where high values indicate more dense vegetation and values near -1 represents water or barren land [21]. NDVI is often associated with rainfall patterns, it can indicate regions with high rainfall and then suitable breeding sites for mosquitoes. NDVI influences both the birth rate and mortality rate of mosquitoes and thus, has great influence on the mosquito density [22].

Malaria transmission models have a crucial role in understanding the transmission dynamics of the diseases [23–25]. These models, in their different forms, have been utilized to identify the most effective intervention strategies for eradicating the disease [26]. Models on vectors have been shown that The initial fundamental malaria model, introduced by Ross [27], was a simplistic susceptible-infective-susceptible (SIS) model. It aimed to describe the link between the mosquito population and the incidence of malaria in humans. The study showed that malaria persistence is subject to certain mosquito population thresholds. Several authors have proposed many improvements to the model by Ross [28–31]. In [31], the authors assumed that the population of infectious mosquitoes is constant, and it can be shown that lowering the number of mosquitoes will not reduce malaria disease. The results showed that the mosquito's lifespan is a critical factor in the transmission of malaria. Parham and Michael [13] developed a model for the effect of temperature and precipitation on the adult emergence rate of mosquitoes, which we will use in our research to determine the impact of weather on the reproduction number in terms of temperature, rainfall, and extend the work by including the vegetation index. In [13], demographic factors have been considered in order to predict the number of fatalities that may occur as a result of illnesses. They also discovered that the influence of temperature on vector abundance had a strong physiological basis, suggesting that the deterministic population model may be significantly favored. Understanding and quantifying the relationship between anopheles mortality and NDVI, can lead to gaining insights into how environmental factors influence mosquito populations and potentially inform strategies for mosquito control and disease prevention.

This study introduces a model for malaria transmission, which assesses the impact of temperature, rainfall, and NDVI on the dynamics of malaria transmission in Burundi. It focuses on how these factors influence the occurrence of malaria, the study offers insights into future malaria trends by utilizing the reproduction number as a predictive tool. The disease transmission dynamics are modeled using non-linear ordinary differential equations (ODE's) where humans and mosquitoes infect each other.

This work is structured as follows: Sect. 2 presents an overview of the model formulation, its epidemiological validity, and the domain in which it is mathematically well posed. We analyze the existence of equilibria and provide calculations for the fundamental reproduction number, as well as an assessment of the stability of the equilibria. Additionally, we describe the study area and the data processing. In Sect. 3, we conduct numerical simulations of the model and forecast temporal trends in the reproduction number and observations. We finally discuss the results obtained from our analysis.

2 Materials and Methods

2.1 Model Formulation

The total human population is subdivided into four classes. The susceptible class $S_H(t)$, those who are exposed to malaria parasites but not infectious $E_H(t)$, individuals with malaria symptoms $I_H(t)$ and the class of recovered individuals $R_H(t)$ to get an SEIR model. We assume that temporary immunity is gained when one has recovered from the disease and has immunity against the malaria parasite for a certain period before losing the immunity.

Susceptible individuals are introduced into the population at a birth rate α_h and acquire the malaria parasite through a bite of an infectious mosquito. They progress to the exposed class through a force of infection governed by the function $\lambda_h = \frac{\beta_1 a(T) I_V(t)}{N_H(t)}$, defined as the rate at which susceptible humans become infected by infectious anopheles mosquitoes. The expression $a(T)$ is the biting rate of the mosquitoes, and β_1 represents the probability that a susceptible individual can be infected by an infectious mosquito, with $0 < \beta_1 \leq 1$. The susceptible individuals also recruit recovered individual from $R_H(t)$, those are individuals who lose their immunity at a rate q . They also die due to a natural death rate $\mu_h(t)$. The rate at which the susceptible individuals changes over time is determined by

$$\frac{dS_H}{dt} = \alpha_h + qR_H - \lambda_h S_H - \mu_h S_H. \quad (1)$$

After being bitten by infectious mosquitoes, susceptible individuals progress to the exposed class through the rate λ_h . This class is exposed to the malaria parasite for a few days before becoming infectious and moving to the infectious class. The number of individuals in the exposed class can also decrease due to the natural mortality rate $\mu_h(t)$. Thus, the rate of change of exposed individuals over time is given by

$$\frac{dE_H}{dt} = \lambda_h S_H - (\rho_h + \mu_h) E_H. \quad (2)$$

Exposed individuals transition to the infectious class at a rate denoted by ρ_h . Once infected, they either recover at a rate γ_h or succumb to the disease, with mortality occurring at rates δ and μ_h , respectively. Here, the parameter δ represents the death rate attributed specifically to the disease. Thus, we obtain the differential equation

$$\frac{dI_H}{dt} = \rho_h E_H - (\mu_h + \gamma_h + \delta) I_H. \quad (3)$$

Individuals who are infectious transition to the recovered class $R_H(t)$ at a rate denoted by γ_h , and those who have recovered lose their immunity at a rate of q . They also die due to natural mortality rate $\mu_h(t)$

$$\frac{dR_H}{dt} = \gamma_h I_H - (\mu_h + q) R_H. \quad (4)$$

The total mosquito populations $N_V(t)$ are subdivided into three classes: $S_V(t)$ susceptible, $E_V(t)$ exposed and $I_V(t)$ infectious mosquitoes. Susceptible mosquitoes ($S_V(t)$) are recruited into mosquito population at a rate $\theta_v(T, R, \eta)$. The function $\theta_v(T, R, \eta)$ represents the birth rate of the mosquitoes, we consider it as a function of temperature, rainfall and

NDVI. In [13, 32], the mosquito birth rate is considered as a function of temperature and rainfall and it is defined as

$$\theta_v(T, R) = \frac{N_E S_E(R) S_L(T, R) S_P(R)}{\chi_E + \chi_L(T) + \chi_P}, \quad (5)$$

where

- N_E is the number of eggs laid per adult female mosquito.
- $S_E(R)$, $S_L(T, R)$ and $S_P(R)$, the survival probability of eggs, larvae, and pupae respectively.
- χ_E , $\chi_L(T)$, χ_P , are the duration of eggs, larva, and pupae stages. The duration of larvae stage varies with temperature [32, 33] and can be represented as $\chi_L(T) = \frac{1}{\xi_1 T + \xi_2}$ where ξ_1 and ξ_2 are constant.

The survival probability of eggs, larvae and pupae is a quadratic relationship between their density and rainfall [13].

$$S_E(R) = \left[\frac{4P_E^*}{R_w^2} \right] R(R_w - R), \quad (6)$$

$$S_{L2}(R) = \left[\frac{4P_L^*}{R_w^2} \right] R(R_w - R), \quad (7)$$

$$S_P(R) = \left[\frac{4P_P^*}{R_w^2} \right] R(R_w - R). \quad (8)$$

P_E^* , P_L^* , P_P^* are the maximum survival probability of eggs, larvae, and pupae, which correspond to the maximum rainfall proliferation and R_w is the washout limit.

The authors in [13, 32] also defined the survival probability of larvae as a function of temperature and rainfall, and

$$S_L(T, R) = S_{L1}(T) S_{L2}(R), \quad (9)$$

where $S_{L1}(T)$ and $S_{L2}(R)$ are the survival probability of larvae depending on temperature and rainfall respectively so that

$$S_{L1}(T) = e^{-\alpha T + \beta}, \quad (10)$$

with α and β constants. The quadratic function of $S_{L2}(R)$ is given in the equation (7). Rainfall has two side effects on the survival probability of eggs, larvae and pupae. Either it can raise the frequency of malaria by providing adequate habitat for mosquitoes or decrease the prevalence by flushing away mosquito breeding sites [32].

NDVI can affect the aquatic stages by providing suitable breeding sites. However, the direct influence on eggs is limited as eggs are laid directly in water and are subject to aquatic conditions. Similar to the eggs stage, the impact of NDVI is limited at the pupae stage as pupae are also found at the water's surface. For all these reasons, the larvae stage is highly affected by NDVI primarily by affecting the quality of their breeding habitats [34, 35]. Areas with high NDVI values indicate denser and healthier vegetation which are more likely to provide suitable conditions for mosquito larvae to develop and grow, including cooler temperatures, food sources, and protection from predators. Conversely, areas with lower NDVI values may lead to less favorable conditions, potentially reducing the survival

probability of mosquito larvae [34, 35]. Hence, following all of these ideas, we included the NDVI's effect in the model by formulating a new mathematical function defining the survival probability of mosquito larvae in terms of temperature (T), rainfall (R), and NDVI (η). The mathematical expression may be then represented as

$$\begin{aligned} S_L(T, R, \eta) &= S_{L1}(T)S_{L2}(R)S_{L3}(\eta), \\ &= S_{L1}(T)S_{L2}(R)(1 + H(\eta)). \end{aligned} \quad (11)$$

Here, $H(\eta)$ represents a quadratic function of the NDVI. $H(\eta)$ has its roots at $\eta = \eta_{Min}$ and $\eta = \eta_{Max}$. In equation (11), the term $1 + H(\eta)$ modulates the survival probability of mosquito larvae and

$$H(\eta) = -(\eta - \eta_{Min})(\eta - \eta_{Max}). \quad (12)$$

The use of NDVI as a factor reflects an ecological understanding of the relationship between vegetation, environmental conditions, and mosquito larvae survival. The formulation is consistent with the idea that NDVI serves as an indicator of habitat quality for mosquitoes during the larvae stage [36].

The mosquito birth rate (5) is then defined in terms of temperature, rainfall, and NDVI as

$$\begin{aligned} \theta_v(T, R, \eta) &= \frac{N_E S_E(R) S_L(T, R, \eta) S_P(R)}{\chi_E + \chi_L(T) + \chi_P}, \\ &= \frac{64 N_E P_E^* P_L^* P_P^* R^3 (R_W - R)^3 (e^{-\alpha T + \beta}) (1 - (\eta - \eta_{Min})(\eta - \eta_{Max}))}{R_W^6 (\chi_E + \chi_L(T) + \chi_P)}. \end{aligned} \quad (13)$$

The reduction in the number of individuals in the class of susceptible mosquitoes is due to the force of infection governed by the function $\lambda_v = \frac{\beta_2 a(T) I_H(t)}{N_H(t)}$ which occurs when there is a bite from susceptible mosquito to an infectious human population $I_H(t)$ and to the natural mortality rate of mosquitoes $\mu_v(T, \eta)$. The natural mortality rate is then considered as a function of temperature and vegetation index in this model, where the formulation of $\mu_v(T, \eta)$ is presented in (20). Hence, susceptible mosquitoes become exposed at λ_v rate. The parameter λ_v denotes the rate at which susceptible mosquitoes are getting infected by infectious human population and β_2 is the probability that an infected human will infect a susceptible mosquito. The rate of change of susceptible mosquitoes over time is

$$\frac{dS_V}{dt} = \theta_v(T, R, \eta) - (\mu_v(T, \eta) + \lambda_v) S_V. \quad (14)$$

After a period t , the exposed mosquitoes progress to the class of infected mosquitoes at a rate ρ_v and die due to natural death at a rate $\mu_v(T, \eta)$. Hence,

$$\frac{dE_V}{dt} = \lambda_v S_V - (\mu_v(T, \eta) + \rho_v) E_V. \quad (15)$$

The exposed mosquitoes progress to the class of infected mosquitoes $I_V(t)$ and the class of infected mosquitoes decreases due to the natural death of mosquitoes and

$$\frac{dI_V}{dt} = \rho_v E_V - \mu_v(T, \eta) I_V. \quad (16)$$

As NDVI increases, it indicates healthier vegetation, which could provide more breeding sites and resources for mosquitoes. This might lead to an increase in mosquito populations, potentially resulting in lower mortality rates, then the relationship between Anopheles mosquito's mortality and NDVI can be represented in a functional form which demonstrates slow mortality decrease as NDVI values increase [22]. As a result, the mathematical function defining the mortality rate $\mu_v(T, \eta)$ is

$$\mu_v(T, \eta) = p(T) * q(\eta), \quad (17)$$

where $p(T)$ and $q(\eta)$ account for the effect of temperature and NDVI respectively on the mortality of mosquitoes. We have that

$$p(T) = [C_T(T - T_M^*)^2 + d], \quad (18)$$

and

$$q(\eta) = \left[\frac{C_\eta}{C_\eta + \alpha_\eta \eta} \right], \quad (19)$$

with C_T being the amplitude of the mortality function and controlling how much mortality rate can vary with temperature, T_M^* is the minimum temperature at which mosquitoes are most likely to survive and have the lowest mortality rate [33], d is the minimum value of the function, C_η is the saturation parameter and α_η is a constant. This yields,

$$\mu_v(T, \eta) = [C_T(T - T_M^*)^2 + d] \left[\frac{C_\eta}{C_\eta + \alpha_\eta \eta} \right] = \frac{C_\eta(d + C_T(T - T_M^*)^2)}{C_\eta + \alpha_\eta \eta}. \quad (20)$$

We denote the total population for humans and mosquitoes by $N_H(t) = S_H(t) + E_H(t) + I_H(t) + R_H(t)$ and $N_V(t) = S_V(t) + E_V(t) + I_V(t)$ respectively. Furthermore, we assume that all newborn humans and mosquitoes are susceptible to the infection. We also assume that infected mosquitoes remain infected all their life [37]. We illustrate the model system in Fig. 1.

The dynamics of transmission of malaria between humans and mosquitoes are then described by the following system of nonlinear ordinary differential equations.

$$\begin{cases} \frac{dS_H}{dt} = \alpha_h + qR_H - \lambda_h S_H - \mu_h S_H, \\ \frac{dE_H}{dt} = \lambda_h S_H - (\rho_h + \mu_h) E_H, \\ \frac{dI_H}{dt} = \rho_h E_H - (\mu_h + \gamma_h + \delta) I_H, \\ \frac{dR_H}{dt} = \gamma_h I_H - (\mu_h + q) R_H, \\ \frac{dS_V}{dt} = \theta_v(T, R, \eta) - (\mu_v(T, \eta) + \lambda_v) S_V, \\ \frac{dE_V}{dt} = \lambda_v S_V - (\mu_v(T, \eta) + \rho_v) E_V, \\ \frac{dI_V}{dt} = \rho_v E_V - \mu_v(T, \eta) I_V, \end{cases} \quad (21)$$

with the following initial conditions:

$$S_H(0) = S_H^0 > 0, E_H(0) = E_H^0 \geq 0, I_H(0) = I_H^0 \geq 0, R_H(0) = R_H^0 \geq 0,$$

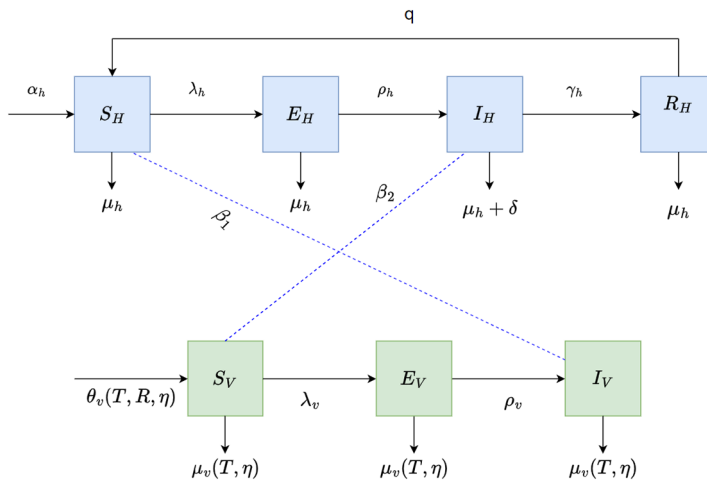


Fig. 1 Flow diagram for the transmission dynamics of malaria. The diagram shows the compartmental relations between the human host and mosquito populations

$$S_V(0) = S_V^0 > 0, E_V(0) = E_V^0 \geq 0, I_V(0) = I_V^0 \geq 0. \quad (22)$$

The variables and parameters used in the model are described in Table 1.

2.2 Model Properties

In this section, we carry out mathematical model analysis without incorporating parameters that rely on climate conditions. We show that the model is mathematically and biologically meaningful. To do this, we define $\theta_v(T, R, \eta)$, $\mu_v(T, \eta)$, and $a(T)$ as functions with maximum values represented by θ_v , μ_v , and a , respectively and, $\theta_v(T, R, \eta) \leq \theta_v$, $\mu_v(T, \eta) \leq \mu_v$, and $a(T) \leq a$.

2.2.1 Feasible Region

Theorem 1 (Positivity) *The solutions $S_H(t)$, $E_H(t)$, $I_H(t)$, $R_H(t)$, $S_V(t)$, $E_V(t)$, $I_V(t)$ of the model (21) are non-negative for all $t \geq 0$ given the initial conditions (22).*

Proof Given that the initial values of the model's variable are nonnegative for $t \geq 0$, we first show that $S_H(t)$ is positive. We assume that there exists an initial time t_1 such that $S_H(t_1) = 0$, $S'_H(t_1) \leq 0$ and

$$E_H(t_1) > 0, I_H(t_1) > 0, R_H(t_1) > 0, S_V(t_1) > 0, E_V(t_1) > 0, I_V(t_1) > 0. \quad (23)$$

We have $\left. \frac{dS_H}{dt} \right|_{t=t_1} = \alpha_h + qR_H(t_1) > 0$ which is a contradiction to our assumption, hence $S_H(t)$ remains non-negative for all $t \geq 0$.

Secondly, we show that $E_H(t)$ is also positive for all $t \geq 0$. Suppose that there exist an initial time t_2 such that $E_H(t_2) = 0$, $E'_H(t_2) < 0$ and

$$S_H(t_2) > 0, I_H(t_2) > 0, R_H(t_2) > 0, S_V(t_2) > 0, E_V(t_2), I_V(t_2) > 0. \quad (24)$$

Table 1 Description of states variables and parameters of the basic malaria model

Variable	Description	Unit
S_H	Susceptible Humans population	people
E_H	Exposed humans population	people
I_H	Infected humans population	people
R_H	Recovered humans population	people
S_V	Susceptible mosquito population	mosquitoes
E_V	Exposed mosquito population	mosquitoes
I_V	Infected mosquito population	mosquitoes
Parameter	Description	Unit
α_h	Recruitment rate of human populations	people/time
q	Rate of loss of immunity	time ⁻¹
β_1	Probability of transmission of malaria from an infectious vector to a susceptible human per contact.	dimensionless
$a(T)$	Mosquito biting rate	time ⁻¹
μ_h	Natural mortality rate of human population	time ⁻¹
$\mu_v(T, \eta)$	Mortality rate of mosquitoes	time ⁻¹
ρ_h	Progression rate of human population from E_H to I_H	time ⁻¹
γ_h	Progression rate of human population from I_H to R_H	time ⁻¹
δ	Disease induced death rate	time ⁻¹
$\theta_v(T, R, \eta)$	Recruitment rate of mosquitoes	mosquito/time
β_2	Probability of transmission of malaria from an infectious human to a susceptible vector per contact.	dimensionless
ρ_v	Progression rate of mosquito population from E_V to I_V	time ⁻¹

We have $\left. \frac{dE_H}{dt} \right|_{t=t_2} = \lambda_h(t_2)S_H(t_2) > 0$ which is a contradiction to our assumption, hence $E_H(t)$ remains non-negative for all $t \geq 0$. Using a similar approach, it can be shown that the model's other state variables remain positive for all $t \geq 0$. $\square$

Theorem 2 (*Boundedness*) *Given the initial values of the variables of the model system (21) that are $S_H^0 > 0$, $E_H^0 \geq 0$, $I_H^0 \geq 0$, $R_H^0 \geq 0$, $S_V^0 > 0$, $E_V^0 \geq 0$, $I_V^0 \geq 0$. The solution of the model (21) are uniformly bounded within the region*

$$\Omega = \{(S_H(t), E_H(t), I_H(t), R_H(t), S_V(t), E_V(t), R_V(t))$$

$$\in \mathbb{R}_+^7 | 0 \leq N_H(t) \leq \frac{\alpha_h}{\mu_h}, N_V(t) \leq \frac{\theta_v}{\mu_v}\}.$$

Proof To prove that the solution of model system (21) are uniformly bounded within Ω let assume that $((S_H(t), E_H(t), I_H(t), R_H(t), S_V(t), E_V(t), I_V(t)))$ is the solution of the model. If $N_H(t) = S_H(t) + E_H(t) + I_H(t) + R_H(t)$ and $N_V(t) = S_V(t) + E_V(t) + I_V(t)$ then

$$\begin{aligned} \frac{dN_H(t)}{dt} &= \alpha - \mu_h N_H(t) - \delta I_H(t) \\ &\leq \alpha - \mu_h N_H(t), \quad \text{since } I_H(t) \geq 0, \end{aligned} \quad (25)$$

and

$$\frac{dN_V(t)}{dt} = \theta_v - \mu_v N_V(t). \quad (26)$$

Using the integrating factor to solve the differential inequality (25) and equation (26) we found that

$$N_H(t) \leq \frac{\alpha_h}{\mu_h} + \left(N_H^0 - \frac{\alpha_h}{\mu_h} \right) e^{-\mu_h t}, \quad (27)$$

$$N_V(t) = \frac{\theta_v}{\mu_v} + \left(N_V^0 - \frac{\theta_v}{\mu_v} \right) e^{-\mu_v t}. \quad (28)$$

If

- $N_H(t) = S_H(t) + E_H(t) + I_H(t) + R_H(t)$ then, $\lim_{t \rightarrow \infty} \sup N_H(t) \leq \frac{\alpha_h}{\mu_h}$.
- $N_V(t) = S_V(t) + E_V(t) + I_V(t)$ then, $\lim_{t \rightarrow \infty} \sup N_V(t) \leq \frac{\theta_v}{\mu_v}$.

As $t \rightarrow \infty$,

$$N_H \text{ approaches } \frac{\alpha_h}{\mu_h},$$

and

$$N_V = \frac{\theta_v}{\mu_v}.$$

Then,

$$N_H(t) = S_H(t) + E_H(t) + I_H(t) + R_H(t) \leq \frac{\alpha_h}{\mu_h},$$

and

$$N_V = S_V(t) + E_V(t) + I_V(t) = \frac{\theta_v}{\mu_v}.$$

At $t = 0$,

$$N_H(0) \geq 0 \text{ and } N_V(0) \geq 0.$$

Therefore, $0 \leq N_H(t) \leq \frac{\alpha_h}{\mu_h}$, $0 \leq N_V(t) \leq \frac{\theta_v}{\mu_v}$ and since all solutions in $\mathbb{R}_+^7$ are uniformly bounded, Ω is a feasible region for the model (21). $\square$

2.2.2 Model Analysis

Here, we examine and analyze the model stability of equilibrium points.

Disease-Free Equilibrium (DFE) Points The DFE exists when there is no disease in the human and mosquito populations. For the model system under analysis, the malaria free equilibrium point E_0 is

$$E_0 = \left(\frac{\alpha_h}{\mu_h}, 0, 0, 0, \frac{\theta_v}{\mu_v}, 0, 0 \right). \quad (29)$$

Reproduction Number We use the next-generation matrix approach described in [38, 39] to find the reproduction number R_0 . The reproduction number is defined as the average number of secondary infections caused by a single infectious individual in a population of susceptible individuals. The basic reproduction number determines how severe a disease can become, whether it can lead to an epidemic or pandemic, or if it will die out. It can also be considered as a measure of the strength of transmission intensity for the disease [40], and has been estimated from both mathematical models and data by many researchers over the years [41, 42]

It can be shown that if $R_0 > 1$, an epidemic occurs which suggests that the disease can sustain transmission in the population. On the other hand, if $R_0 < 1$ then no epidemic is likely to occur. Consequently, reducing R_0 below 1 is key to malaria control efforts to prevent and manage outbreaks.

The NGM is the matrix product of the new infection matrix (F) and the inverse of the transition matrix (V). The spectral radius (ρ) of the NGM is the dominant eigenvalue, which is equal to R_0 .

Let f represents the rate of new infections, while v denotes the transition terms in our model. Their corresponding Jacobian matrices, F and V , are derived by linearizing the system around the DFE E_0 .

Specifically, we have:

$$f = \begin{pmatrix} \lambda_h S_H \\ 0 \\ \lambda_v S_V \\ 0 \end{pmatrix}; \quad v = \begin{pmatrix} (\rho_h + \mu_h)E_H \\ (\mu_h + \delta + \gamma_h)I_H - \rho_h E_H \\ (\mu_v + \rho_v)E_V \\ \mu_v I_V - \rho_v E_V \end{pmatrix}. \quad (30)$$

F and V are given by:

$$F = \begin{pmatrix} 0 & 0 & 0 & \beta_1 a \\ 0 & 0 & 0 & 0 \\ 0 & \frac{\beta_2 a \theta_v \mu_h}{\mu_v \alpha_h} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ and } V = \begin{pmatrix} \rho_h + \mu_h & 0 & 0 & 0 \\ -\rho_h & \mu_h + \delta + \gamma_h & 0 & 0 \\ 0 & 0 & \mu_v + \rho_v & 0 \\ 0 & 0 & -\rho_v & \mu_v \end{pmatrix} \quad (31)$$

The inverse of V is:

$$V^{-1} = \begin{pmatrix} \frac{1}{\rho_h + \mu_h} & 0 & 0 & 0 \\ \frac{\rho_h}{(\rho_h + \mu_h)(\mu_h + \delta + \gamma_h)} & \frac{1}{\mu_h + \delta + \gamma_h} & 0 & 0 \\ 0 & 0 & \frac{1}{\mu_v + \rho_v} & 0 \\ 0 & 0 & \frac{\rho_v}{\mu_v(\mu_v + \rho_v)} & \frac{1}{\mu_v} \end{pmatrix}, \quad (32)$$

$$FV^{-1} = \begin{pmatrix} 0 & 0 & \frac{\beta_1 a \rho_v}{\mu_v b_3} & \frac{\beta_1 a}{\mu_v} \\ 0 & 0 & 0 & 0 \\ \frac{\beta_2 a \rho_h \theta_v \mu_h}{\mu_v \alpha_h b_1 b_2} & \frac{\beta_2 a \theta_v \mu_h}{\mu_v \alpha_h b_2} & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (33)$$

and,

$$R_0 = \rho(FV^{-1}) = \sqrt{\frac{\beta_1 \beta_2 a^2 \rho_h \rho_v \theta_v \mu_h}{\mu_v^2 \alpha_h b_1 b_2 b_3}} = \sqrt{R_{0h} * R_{0v}} \quad (34)$$

where

$$b_1 = \rho_h + \mu_h, b_2 = \mu_h + \gamma_h + \delta, b_3 = \mu_v + \rho_v.$$

and

$$R_{0h} = \frac{\beta_1 a \rho_h \mu_h}{\alpha_h b_1 b_2}, R_{0v} = \frac{\beta_2 a \rho_v \theta_v}{\mu_v^2 b_3}$$

Local Stability of the DFE The local stability of the disease free equilibrium point (DFE) can be shown through the linearized form of (21) evaluated at E_0 .

Theorem 3 The DFE E_0 is locally asymptotically stable if $R_0 < 1$ and unstable if $R_0 > 1$

Proof The Jacobian matrix of the model at E_0 is given by

$$J(E_0) = \begin{pmatrix} -\mu_h & 0 & 0 & q & 0 & 0 & -\beta_1 a \\ 0 & -b_1 & 0 & 0 & 0 & 0 & \beta_1 a \\ 0 & \rho_h & -b_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma_h & -(\mu_h + q) & 0 & 0 & 0 \\ 0 & 0 & \frac{-\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & 0 & -\mu_v & 0 & 0 \\ 0 & 0 & \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & 0 & 0 & -b_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_v & -\mu_v \end{pmatrix} \quad (35)$$

The Jacobian matrix has two columns with diagonal entries. Hence, $-\mu_h$ and $-\mu_v$ are two of the eigenvalues of the matrix (35) with negative real parts. To find others let's reduce the Jacobian matrix by eliminating the first and fifth columns and the corresponding rows to these diagonal entries. We find

$$J_1(E_0) = \begin{pmatrix} -b_1 & 0 & 0 & 0 & \beta_1 a \\ \rho_h & -b_2 & 0 & 0 & 0 \\ 0 & \gamma_h & -(\mu_h + q) & 0 & 0 \\ 0 & \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & 0 & -b_3 & 0 \\ 0 & 0 & 0 & \rho_v & -\mu_v \end{pmatrix} \quad (36)$$

Using the same approach for the third column we find another eigenvalue $-(\mu_h + q)$ with negative real parts and (36) becomes:

$$J_2(E_0) = \begin{pmatrix} -b_1 & 0 & 0 & \beta_1 a \\ \rho_h & -b_2 & 0 & 0 \\ 0 & \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & -b_3 & 0 \\ 0 & 0 & \rho_v & -\mu_v \end{pmatrix} \quad (37)$$

To check if the remaining eigenvalues have the negative real parts and ensure the local stability when $R_0 < 1$, we will use Routh-Hurwitz conditions to see if they are sufficient and necessary [43]. The polynomial characteristic equation of (37) is

$$\lambda^4 + (b_1 + b_2 + b_3 + \mu_v)\lambda^3 + (b_1b_2 + b_1b_3 + b_2b_3 + b_1\mu_v + b_2\mu_v + b_3\mu_v)\lambda^2 + (b_1b_2b_3 + b_1b_2\mu_v + b_1b_3\mu_v + b_2b_3\mu_v)\lambda + b_1b_2b_3\mu_v - \frac{\beta_1\beta_2a^2\rho_v\rho_h\theta_v\mu_h}{\alpha_h\mu_v} = 0 \quad (38)$$

We set

$$\begin{aligned} J_3 &= b_1 + b_2 + b_3 + \mu_v, \\ J_2 &= b_1b_2 + b_1b_3 + b_2b_3 + b_1\mu_v + b_2\mu_v + b_3\mu_v, \\ J_1 &= b_1b_2b_3 + b_1b_2\mu_v + b_1b_3\mu_v + b_2b_3\mu_v, \\ J_0 &= b_1b_2b_3\mu_v - \frac{\beta_1\beta_2a^2\rho_v\rho_h\theta_v\mu_h}{\alpha_h\mu_v}, \\ &= b_1b_2b_3\mu_v(1 - R_0^2), \end{aligned}$$

where $J_3 > 0$, $J_2 > 0$, $J_1 > 0$ and $J_0 > 0$ iff $R_0 < 1$. The Hurwitz determinants are given by:

$$\begin{aligned} \det(H_1) &= J_3 = b_1 + b_2 + b_3 + \mu_v > 0, \\ \det(H_2) &= \begin{vmatrix} J_3 & 1 \\ J_1 & J_2 \end{vmatrix} = J_3J_2 - J_1 \\ &= b_1^2(b_2 + b_3 + \mu_v) + b_2^2(b_1 + b_3 + \mu_v) + b_3^2(b_1 + b_2 + \mu_v) + \mu_v^2(b_1 + b_2 + b_3) \\ &\quad + 2b_1b_2b_3 + 2\mu_v(b_1b_2 + b_1b_3 + b_2b_3) > 0 \\ \det(H_3) &= \begin{vmatrix} J_3 & 1 & 0 \\ J_1 & J_2 & J_3 \\ 0 & J_0 & J_1 \end{vmatrix} = J_1(J_3J_2 - J_1) - J_3^2J_0 \\ &= (b_1 + b_2 + b_3 + \mu_v)[(b_1^2(b_2 + b_3 + \mu_v) + b_2^2(b_1 + b_3 + \mu_v) + b_3^2(b_1 + b_2 + \mu_v) \\ &\quad + \mu_v^2(b_1 + b_2 + b_3) + 2\mu_v(b_1b_2 + b_1b_3 + b_2b_3) + 2b_1b_2b_3) \\ &\quad + ((b_1 + b_2 + b_3 + \mu_v)(b_1b_2b_3\mu_v(1 - R_0^2)))] > 0 \quad \text{iff } R_0 < 1 \\ \det(H_4) &= \begin{vmatrix} J_3 & 1 & 0 & 0 \\ J_1 & J_2 & J_3 & 1 \\ 0 & J_0 & J_1 & J_2 \\ 0 & 0 & 0 & J_0 \end{vmatrix} = J_0H_3 > 0. \end{aligned}$$

All of the determinants of the Hurwitz matrices are positive implying that all the eigenvalues have a negative real part. Thus, the DFE E_0 is local asymptotically stable (LAS) when $R_0 < 1$ and unstable $R_0 > 1$. $\square$

Global Stability of the DFE The Global stability of the DFE can be established using the approach described by Chavez in [44]. As the DFE is locally asymptotically stable whenever $R_0 < 1$ and unstable when $R_0 > 1$, we provide two requirements that must be satisfied to ensure that the disease-free state is globally asymptotically stable.

Thus, we re-write (21) in the form

$$\begin{cases} \frac{dX}{dt} = F(X, Y) \\ \frac{dY}{dt} = G(X, Y), \quad G(X, 0) \end{cases} \quad (39)$$

where $X \in \mathbb{R}_+^2$ represents the number of susceptible individuals and susceptible mosquitoes and $Y \in \mathbb{R}_+^4$ is the number of exposed humans, infected humans, exposed mosquitoes, infected mosquitoes. $U_0 = (u_0, 0)$ denotes the DFE. To guarantee the global stability of the DFE, the conditions (A_1) and (A_2) below must be satisfied:

(A_1) For $\frac{dX}{dt} = F(x, 0)$, U_0 is globally asymptotically stable (GAS).

(A_2) $G(X, Y) = AY - \hat{G}(X, Y)$, $\hat{G}(X, Y) \geq 0$ for $(X, Y) \in \Omega$ the feasible region of the model.

$A = \frac{dG(u_0, 0)}{dy}$ is a M -matrix. A M -matrix is a matrix with all the off-diagonal elements that are non-negative and the real part of every one of its eigenvalues is greater than or equal to zero.

Corollary 1 [44] *The equilibrium point u_0 of (39) is GAS equilibrium if $R_0 \leq 1$.*

Theorem 4 *The steady-state E_0 is GAS if $R_0 < 1$*

Proof $X = (S_H, S_V)^T$, $Y = (E_H, I_H, E_V, I_V)^T$ and, $u_0 = \left(\frac{\alpha_h}{\mu_h}, \frac{\theta_v}{\mu_v} \right)$.

For the first requirement, from (27) and (28) as $t \rightarrow \infty$, we have $S_H(t) \rightarrow \frac{\alpha_h}{\mu_h}$ and $S_V(t) \rightarrow \frac{\theta_v}{\mu_v}$.

Then, u_0 is a globally asymptotically equilibrium point as the solution of $\frac{dS_H(t)}{dt}$ and $\frac{dS_V(t)}{dt}$ converge to this point.

$$\frac{dX}{dt} = F(X, 0) = \begin{pmatrix} \alpha_h - \mu_h S_H \\ \theta_v - \mu_v S_V \end{pmatrix}, \quad (40)$$

and

$$\frac{dY}{dt} = G(X, Y) = \begin{pmatrix} \lambda_h S_H - (\rho_h + \mu_h) E_H \\ \rho_h E_H - (\mu_h + \gamma_h + \delta) I_H \\ \lambda_v S_V - (\mu_v + \rho_v) E_V \\ \rho_v E_V - \mu_v I_V \end{pmatrix}, \quad (41)$$

$$A = \frac{dG(u_0, 0)}{dt} = \begin{pmatrix} -(\rho_h + \mu_h) & 0 & 0 & \beta_1 a \\ \rho_h & -(\mu_h + \gamma_h + \delta) & 0 & 0 \\ 0 & \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & -(\mu_v + \rho_v) & 0 \\ 0 & 0 & \rho_v & -\mu_v \end{pmatrix}, \quad (42)$$

Clearly A is a M -matrix and $\hat{G}(X, Y) = AY - G(X, Y)$
 So,

$$\hat{G}(X, Y) = \begin{pmatrix} \hat{G}_1(X, Y) \\ \hat{G}_2(X, Y) \\ \hat{G}_3(X, Y) \\ \hat{G}_4(X, Y) \end{pmatrix} = \begin{pmatrix} \beta_1 a I_V (1 - \frac{S_H}{N_H}) \\ 0 \\ \beta_2 a I_H (\frac{S_V^0}{S_H^0} - \frac{S_V}{N_H}) \\ 0 \end{pmatrix} \geq \begin{pmatrix} \beta_1 a I_V (1 - \frac{S_H}{N_H}) \\ 0 \\ \frac{\beta_2 a I_H}{S_H^0} (S_V^0 - S_V) \\ 0 \end{pmatrix} \geq 0 \quad (43)$$

Note that $\beta_1 a I_V (1 - \frac{S_H}{N_H}) \geq 0$ since $0 < S_H \leq \frac{\alpha_h}{\mu_h}$ and $(S_V^0 - S_V) \geq 0$ since $S_V \leq N_V \leq S_V^0 = \frac{\theta_v}{\mu_v}$ and as a result, the conditions (A_1) and (A_2) are met. The global stability of the DFE is proven. $\square$

Existence of the Endemic Equilibrium Point The endemic equilibrium point EE^* is defined as the steady solution when the disease persists in the population. For the model (21) the endemic equilibrium point is given by $EE^* = (S_H^*, E_H^*, I_H^*, R_H^*, S_V^*, E_V^*, I_V^*)$ computed by setting the right-hand sides of the model to zero, the equations of the system (21) are solved at the endemic equilibrium points in terms of the force of infections λ_h^* and λ_v^* to obtain

$$\begin{cases} S_H^* = \frac{\alpha_h (\mu_h + \rho_h) (\mu_h + q) (\delta + \gamma_h + \mu_h)}{[(\mu_h + q)(\lambda_h^* + \mu_h)(\mu_h + \gamma_h + \delta)(\rho_h + \mu_h) - q\gamma_h \lambda_h^* \rho_h]}, \\ E_H^* = \frac{\alpha_h \lambda_h^* (\mu_h + q) (\delta + \gamma_h + \mu_h)}{[(\mu_h + q)(\lambda_h^* + \mu_h)(\mu_h + \gamma_h + \delta)(\rho_h + \mu_h) - q\gamma_h \lambda_h^* \rho_h]}, \\ I_H^* = \frac{\alpha_h \lambda_h^* \rho_h (\mu_h + q)}{[(\mu_h + q)(\lambda_h^* + \mu_h)(\mu_h + \gamma_h + \delta)(\rho_h + \mu_h) - q\gamma_h \lambda_h^* \rho_h]}, \\ R_H^* = \frac{\theta_v}{[(\mu_h + q)(\lambda_h^* + \mu_h)(\mu_h + \gamma_h + \delta)(\rho_h + \mu_h) - q\gamma_h \lambda_h^* \rho_h]}, \\ S_V^* = \frac{\lambda_v^* + \mu_v}{\lambda_v^* \theta_v}, \\ E_V^* = \frac{\lambda_v^* \theta_v}{(\lambda_v^* + \mu_v)(\mu_v + \rho_v)}, \\ I_V^* = \frac{\mu_v (\lambda_v^* + \mu_v)}{\lambda_v^* \rho_v \theta_v}, \\ N_H^* = \frac{\alpha_h [(\rho_h + \mu_h)(\mu_h + \gamma_h + \delta)(\mu_h + q) + \lambda_h^* (\rho_h + \mu_h)(\mu_h + q) + \lambda_h^* \rho_h (\mu_h + q) + \lambda_h^* \rho_h \gamma_h]}{[(\mu_h + q)(\lambda_h^* + \mu_h)(\mu_h + \gamma_h + \delta)(\rho_h + \mu_h) - q\gamma_h \lambda_h^* \rho_h]}. \end{cases} \quad (44)$$

Substituting λ_v^* , I_V^* and N_H^* into the expression of the force of infection λ_h^* we obtain an equation for λ_h^*

$$g(\lambda_h^*) = \lambda_h^* (\varphi_2 \lambda_h^{*2} + \varphi_1 \lambda_h^* + \varphi_0), \quad (45)$$

where

$$\begin{aligned} \varphi_2 &= b_3 \alpha_h \mu_v (b_4 (b_2 + \rho_h) + \rho_h) (\rho_h (b_4 (a\beta_2 + \mu_v) + \mu_v) + b_2 b_4 \mu_v), \\ \varphi_1 &= a^2 \beta_1 \beta_2 b_4 q \gamma_h \rho_h^2 \theta_v \rho_v - a^2 b_1 b_2 \beta_1 \beta_2 b_4^2 \alpha_h \theta_v \rho_v \\ &\quad + a b_1 b_2 b_3 \beta_2 b_4^2 \alpha_h \rho_h \mu_v + 2 b_1 b_2 b_3 b_4^2 \alpha_h \lambda_h \rho_h \mu_v^2 \\ &\quad + 2 b_1 b_2 b_3 b_4 \alpha_h \rho_h \mu_v^2 + 2 b_1 b_2^2 b_3 b_4^2 \alpha_h \mu_v^2, \end{aligned}$$

Table 2 Possible number of positive roots of equation of (46)

φ_2	φ_1	φ_0	Number of positive roots
+	+	+	0
+	+	-	1
+	-	+	2 or 0
+	-	-	1

$$\varphi_0 = b_1^2 b_2^2 b_3 b_4^2 \alpha_h \mu_v^2 (1 - R_0^2).$$

From (45), we get $\lambda_h^* = 0$, which corresponds to the disease free equilibrium point, otherwise

$$\varphi_2 \lambda_h^{*2} + \varphi_1 \lambda_h^* + \varphi_0 = 0, \quad (46)$$

corresponding to the solution of the endemic point. The existence of the endemic equilibria is determined by the number of positive roots of (46).

Solving for λ_h^* , we have two roots

$$\lambda_{h1}^* = \frac{-\varphi_1 + \sqrt{\varphi_1^2 - 4\varphi_2\varphi_0}}{2\varphi_2}, \quad \lambda_{h2}^* = \frac{-\varphi_1 - \sqrt{\varphi_1^2 - 4\varphi_2\varphi_0}}{2\varphi_2}.$$

We note that the coefficients of (46), $\varphi_2 > 0$, φ_1 can be either positive or negative and φ_0 can be positive and negative depending on the value of R_0 , that's if $R_0 < 1$, then $\varphi_0 > 0$ and if $R_0 > 1$, then $\varphi_0 < 0$.

Lemma 1 (i) *The model has a unique endemic equilibrium point if $R_0 > 1$ ($\varphi_0 < 0$),*
(ii) *Two endemic equilibrium points if $R_0 < 1$, and $\Delta = \varphi_1^2 - 4\varphi_2\varphi_0 > 0$,*
(iii) *No endemic equilibrium point otherwise.*

Having R_0 less than one is usually thought to mean that the disease can be eliminated. However, in our model the possibility of the existence of multiple endemic equilibria when $R_0 < 1$ suggests that a backward bifurcation may occur. In epidemic models, a backward bifurcation means that we have two stable situations happening at the same time. So that, even when we have R_0 less than one, the disease can still persist. In these cases, it is crucial to consider not only the R_0 value but also the threshold value of the R_0 parameter that plays a significant role in determining how and when the disease can be controlled or eradicated [45]. We determine the threshold value of the reproduction number, denoted as R_0^c , by solving the equation $\Delta = 0$, we find

$$R_0^c = 1 - \frac{\varphi_1^2}{4\varphi_2 b_1^2 b_2^2 b_3 b_4^2 \alpha_h \mu_v^2}. \quad (47)$$

Bifurcation Analysis We performed the center manifold theory developed by Chavez and Song in [46] to investigate the Bifurcation analysis. This theory is used to investigate the presence of both backward and forward bifurcations. Forward bifurcation occurs when the endemic equilibrium is locally asymptotically stable for $R_0 > 1$ but R_0 approaches 1.

Let us rewrite the variables $S_H, E_H, I_H, R_H, S_V, E_V, I_V$ as $S_H = x_1, E_H = x_2, I_H = x_3, R_H = x_4, S_V = x_5, E_V = x_6, I_V = x_7$ and let $X = (x_1, x_2, x_3, \dots, x_7)^T$, the expressions of X in vector notation. The model system (21) can be rewritten in terms of

$\frac{dX}{dt} = (f_1, f_2, f_3, \dots, f_7)^T$ where

$$\begin{cases} \frac{dx_1}{dt} = \alpha_h + qx_4 - \lambda_h x_1 - \mu_h x_1 = f_1, \\ \frac{dx_2}{dt} = \lambda_h x_1 - (\rho_h + \mu_h)x_2 = f_2, \\ \frac{dx_3}{dt} = \rho_h x_2 - (\mu_h + \gamma_h + \delta)x_3 = f_3, \\ \frac{dx_4}{dt} = \gamma_h x_3 - (\mu_h + q)x_4 = f_4, \\ \frac{dx_5}{dt} = \theta_v - (\mu_v + \lambda_v)x_5 = f_5, \\ \frac{dx_6}{dt} = \lambda_v x_5 - (\mu_v + \rho_v)x_6 = f_6, \\ \frac{dx_7}{dt} = \rho_v x_6 - \mu_v x_7 = f_7, \end{cases} \quad (48)$$

with $\lambda_h = \frac{\beta_1 a x_7}{N_H}$, $\lambda_v = \frac{\beta_2 a x_3}{N_H}$ and $N_H = x_1 + x_2 + x_3 + x_4$, $N_V = x_6 + x_7 + x_8$.

Let us introduce the bifurcation parameter $\beta_1 = \beta_1^*$ when $R_0 = 1$. It follows that

$$\beta_1^* = \beta_1 = \frac{\alpha_h \mu_v (\rho_h + \mu_h)(\mu_h + \delta + \gamma_h)(\mu_v + \rho_v)}{\beta_2 a^2 \rho_h \rho_v \theta_v \mu_h} \quad (49)$$

The Jacobian matrix $J(E_0, \beta_1^*)$ of (48) at the DFE E_0 when $R_0 = 1$ is given by

$$J(E_0, \beta_1^*) = \begin{pmatrix} -\mu_h & 0 & 0 & q & 0 & 0 & -\beta_1^* a \\ 0 & -b_1 & 0 & 0 & 0 & 0 & \beta_1^* a \\ 0 & \rho_h & -b_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \gamma_h & -b_4 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & 0 & -\mu_v & 0 & 0 \\ 0 & 0 & \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} & 0 & 0 & -b_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_v & -\mu_v \end{pmatrix}, \quad (50)$$

with

$$b_1 = \rho_h + \mu_h, b_2 = \mu_h + \delta + \gamma_h, b_3 = \mu_v + \rho_v, b_4 = \mu_h + q.$$

Since the jacobian matrix (50) has a simple zero eigenvalue, we can use the center manifold theory to describe the dynamics of the system near β_1^* when $R_0 = 1$. We compute $w = (w_1, w_2, w_3, \dots, w_8)^T$, the right eigenvector of the jacobian matrix associated with the

zero eigenvalue as follows:

$$\left\{ \begin{array}{ll} -\mu_h w_1 + q w_4 - \beta_1^* a w_7 = 0, & w_1 = \frac{\alpha_h \gamma_h \mu_v^2 q b_3 - \beta_1^* \beta_2 a^2 \mu_h \theta_v \rho_v b_4}{\beta_2 a \mu_h^2 \theta_v \rho_v b_4}, \\ -b_1 w_2 + \beta_1^* a w_7 = 0, & w_2 = \frac{\alpha_h \mu_v^2 b_2 b_3}{\beta_2 a \mu_h \theta_v \rho_v \rho_h}, \\ \rho_h w_2 - b_2 w_3 = 0, & w_3 = \frac{\alpha_h \mu_v^2 b_3}{\beta_2 a \mu_h \theta_v \rho_v}, \\ \gamma_h w_3 - b_4 w_4 = 0, & w_4 = \frac{\alpha_h \gamma_h \mu_v^2 b_3}{\beta_2 a \mu_h \theta_v \rho_v b_4}, \\ -\frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} w_3 - \mu_v w_5 = 0, & w_5 = -\frac{b_3}{\rho_v}, \\ \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} w_3 - b_3 w_6 = 0, & w_6 = \frac{\mu_v}{\rho_v}, \\ \rho_v w_6 - \mu_v w_7 = 0, & w_7 = 1. \end{array} \right. \quad (51)$$

Consider $v = (v_1, v_2, \dots, v_8)$ as a left eigenvector associated with the zero eigenvalue, which satisfies $v \cdot w = 1$. This is determined by solving the system

$$\left\{ \begin{array}{l} -\mu_h v_1 = 0, \\ -b_1 v_2 + \rho_h v_3 = 0, \\ -b_2 v_3 + \gamma_h v_4 - \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} v_5 + \frac{\beta_2 a \theta_v \mu_h}{\alpha_h \mu_v} v_6 = 0, \\ q v_1 - b_4 v_4 = 0, \\ -\mu_v v_5 = 0, \\ -b_3 v_6 + \rho_v v_7 = 0, \\ -\beta_1^* a v_1 + \beta_1^* a v_2 - \mu_v v_7 = 0. \end{array} \right. \quad (52)$$

Clearly, $v_1 = v_4 = v_5 = 0$ and $v \cdot w = 1$ implies that

$$\frac{b_3 v_3 \alpha_h \mu_v^2}{a \beta_2 \mu_h \theta_v \rho_v} + \frac{b_2 b_3 v_2 \alpha_h \mu_v^2}{a \beta_2 \mu_h \rho_h \theta_v \rho_v} + \frac{v_6 \mu_v}{\rho_v} + v_7 = 1 \quad (53)$$

From (52) and (53), we have

$$\left\{ \begin{array}{l} v_1 = 0, v_4 = 0, v_5 = 0 \\ v_2 = \frac{a \beta_2 \mu_h \rho_h \theta_v \mu_v \rho_v}{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v + b_1 b_2 \alpha_h \mu_v^3 + b_1 b_3 \alpha_h \mu_v^3 + b_2 b_3 \alpha_h \mu_v^3}, \\ v_3 = \frac{a b_1 \beta_2 \mu_h \theta_v \mu_v \rho_v}{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v + b_1 b_2 \alpha_h \mu_v^3 + b_1 b_3 \alpha_h \mu_v^3 + b_2 b_3 \alpha_h \mu_v^3}, \\ v_6 = \frac{b_1 b_2 \alpha_h \mu_v^2 \rho_v}{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v + b_1 b_2 \alpha_h \mu_v^3 + b_1 b_3 \alpha_h \mu_v^3 + b_2 b_3 \alpha_h \mu_v^3}, \\ v_7 = \frac{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v}{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v + b_1 b_2 \alpha_h \mu_v^3 + b_1 b_3 \alpha_h \mu_v^3 + b_2 b_3 \alpha_h \mu_v^3}. \end{array} \right. \quad (54)$$

We computed A and B coefficients, in order to determine the existence possibility of a bifurcation

$$A = \sum_{k,i,j=1}^7 v_k w_i w_j \frac{\partial^2 f_k(E_0)}{\partial x_i \partial x_j}, B = \sum_{k,i}^7 v_k w_i \frac{\partial^2 f_k(E_0)}{\partial x_i \partial \beta_1} \quad (55)$$

Since $v_1 = v_4 = v_5 = 0$ and after computing the associated non-zero partial derivatives of f at E_0 , we obtain

$$A = -2v_6 w_1 w_3 \frac{\beta_2 a \theta_v \mu_h^2}{\mu_v \alpha_h^2} + 2v_6 w_3 w_5 \frac{\beta_2 a \mu_h}{\alpha_h} \quad (56)$$

$$B = \frac{a^2 \beta_2 \mu_h \rho_h \theta_v \mu_v \rho_v}{a^2 \beta_1 \beta_2 \mu_h \rho_h \theta_v \rho_v + b_1 b_2 \alpha_h \mu_v^3 + b_1 b_3 \alpha_h \mu_v^3 + b_2 b_3 \alpha_h \mu_v^3} \quad (57)$$

From (57), we clearly see that $B > 0$. To determine the direction of the bifurcation when $R_0 = 1$, we need to know the sign of A . After substituting v_6 , w_1 , w_3 , and w_5 into (56), let us write $A = A_1 - A_2$.

where

$$A_1 = \frac{\beta_1}{\mu_h}, A_2 = \frac{b_3}{\rho_v} + \frac{\alpha_h \gamma_h \mu_v^2 q b_3}{\beta_2 a \mu_h^2 \theta_v \rho_v b_4}$$

- If $A_1 > A_2$, we have $A > 0$ and $B > 0$ then system (21) will undergo a backward bifurcation at $R_0 = 1$.
- If $A_1 < A_2$, we have $A < 0$ and $B > 0$ then system (21) will undergo a forward bifurcation at $R_0 = 1$.

While the conditions $A_1 > A_2$ and $A_1 < A_2$ indicate specific types of bifurcations, it is important to recognize that the precise sign of A cannot be conclusively established based only on the information at hand. If backward bifurcation occurs, the DFE and an endemic equilibrium can coexist even when $R_0 < 1$. Two stable equilibria (the DFE and the endemic state) exist under the same conditions. Whether the system ends up at the DFE or the endemic equilibrium depends on initial conditions, specifically how widespread the disease is at the start.

2.3 Study Area and Data Processing

Burundi, a landlocked country in East Africa, is bordered by Rwanda to the north, Tanzania to the east and south, and the Democratic Republic of the Congo to the west. Positioned between latitudes 2.3°S and 4.5°S, and longitudes 29°E and 31°E, Burundi exhibits a diverse climate, ranging from temperate highlands to tropical lowlands. The data utilized in this study include three main sources. Firstly, the climate data employed for analysis in this study were sourced from the IGEU (Burundi Geographical institute) database, from 2010 to 2022. This dataset, consisting of monthly records, provided crucial information regarding climatic parameters such as temperature and Rainfall. The utilization of data over this time period allowed for a comprehensive examination of long-term climate trends and patterns, enabling a deeper understanding of the dynamics and fluctuations within Burundi. Secondly, there is a collection of time series data derived from the moderate-resolution Imaging Spectroradiometer (Modis). NDVI data was generated using the Google Earth Engine (GEE), a

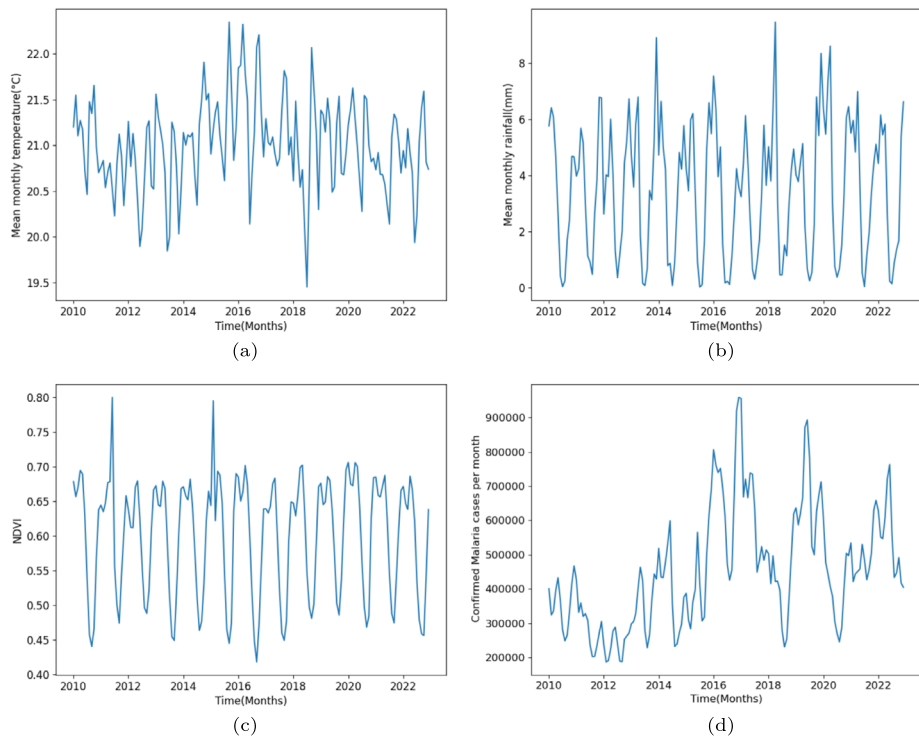
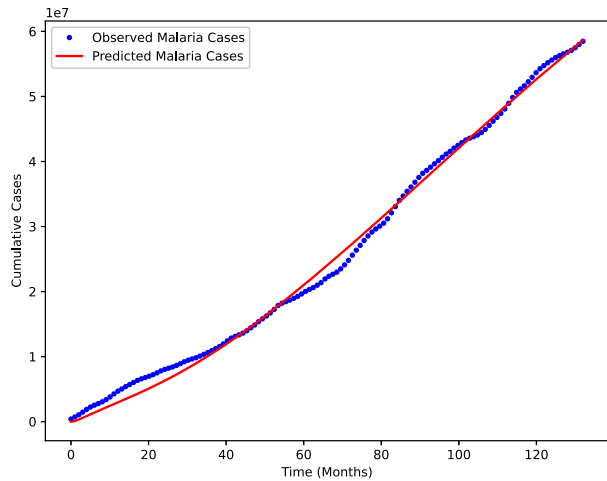


Fig. 2 Country-level mean monthly temperature, mean monthly rainfall, NDVI data, and confirmed malaria cases per month from 2010 to 2022

platform for satellite data collection from the MODIS/006/MCD43A4 surface reflectance composites on a monthly basis with a spatial resolution of 500 meters, spanning from 2010 to 2022 [47]. Figures 2a to 2c display the temperature, rainfall and NDVI distribution across Burundi. The data of confirmed malaria cases used are obtained from the Integrated National Malaria Control Program (INMCP).

Correlation analysis is a statistical technique used to measure the strength and direction of the relationship between two or more variables. In this context, correlation analysis was applied to assess the relationship between NDVI, precipitation, and temperature. The results are developed in the appendix Sect. A. To predict the reproduction number accurately, we utilized forecasting data for temperature, rainfall, and NDVI. This forecasting was conducted using the Seasonal Autoregressive Integrated Moving Average (SARIMA) model [48]. SARIMA is a statistical technique used for time series analysis, particularly for forecasting future values based on historical data. By applying SARIMA to the historical data of temperature, rainfall, and NDVI, we were able to project their future trends, which are essential inputs for predicting the reproduction number in our model (Details are in appendix Sect. B).

Fig. 3 Model fit to data for Malaria in Burundi



3 Numerical Simulation of the Model with Climate-Dependent Parameters

3.1 Parameter Estimation

Estimating the parameters in our model is crucial for conducting effective numerical analysis, ensuring that our results are both meaningful and reliable. This involves determining the specific values of the variables within the model so that when numerical analysis is performed, the outcomes accurately reflect real world conditions. To achieve this, we employed model calibration techniques to establish ideal representation curves for each state variable, aiming to closely approximate their values. In our study, we focus on a scenario where the population of humans in Burundi has stabilized, with a population-to-mortality ratio of $\alpha_h/\mu_h = 12,890,000$, and an average life expectancy of 62.37 years for humans in Burundi. Accordingly, the mortality rate and recruitment rate are calculated as $1/(62.37 * 12)$ and $\alpha_h = 0.0013 * 12,890,000$ per month, respectively [49]. For our simulations, we set initial conditions as follows:

$$S_H(0) = 12,590,000, E_H(0) = 0, I_H(0) = 200,000, R_H(0) = 100,000,$$

$$S_V(0) = 15,000,000, E_V(0) = 0, I_V(0) = 250,000.$$

We applied the least squares method to fit the cumulative incidence equation of the model to the cumulative malaria data from 2010 to 2022 (Fig. 3), initializing the parameters with the initial values derived from the baseline graphs (see Fig. 9).

All parameters used in the simulations are detailed in Table 3.

3.2 Sensitivity Analysis

The Latin Hypercube Sampling method(LHS) and Partial Rank Correlation Coefficients(PRCC) are used to carry out the global sensitivity analysis [51]. Sensitivity analysis is indeed a crucial tool in understanding how uncertainty in model inputs affects the output of a system. Initially, we perform sensitivity analysis using data covering the specified ranges and baseline values for temperature, rainfall, and NDVI-dependent parameters (Table 4).

Table 3 Parameter values and references

Parameter	Value	Reference	Parameter	values	Reference
α_h	17,635	calculated	χ_P	1	[13]
q	$\frac{1}{20 \times 12}$	[50]	T_M^*	16	[33]
β_1	0.45	estimated	P_E	0.93	[32]
μ_h	0.001336	calculated	P_L	0.25	[32]
ρ_h	0.89	estimated	P_P	0.75	[32]
γ_h	0.20	estimated	α	0.05547	[33]
δ	0.005	estimated	β	0.06773	[33]
β_2	0.35	estimated	R_w	50 mm	[13]
ρ_v	0.025	estimated	C_T	0.0009	estimated
N_E	200	[13]	d	0.05	estimated
χ_E	1	[13]	C_η	0.0007	estimated
α_η	0.06	estimated			

Table 4 Temperature, rainfall, and NDVI-dependent parameter ranges

Temperature	Rainfall	NDVI	$\theta_v(T, R, \eta)$	$\mu_v(T, \eta)$	$a(T)$	Scenarios
[16–25 °C]	[5–25 mm]	[0.2–0.4]	$[10.2\text{--}2.16.10^2]$	[0.014–0.035]	[0.07162–0.28074]	(a)
		[0.4–0.8]	$[11.2\text{--}2.16.10^2]$	[0.0093–0.021]	[0.07162–0.28074]	(b)
	[25–50 mm]	[0.2–0.4]	$[0\text{--}1.82.10^2]$	[0.014–0.035]	[0.07162–0.28074]	(c)
		[0.4–0.8]	$[0\text{--}1.71.10^2]$	[0.0093–0.021]	[0.07162–0.28074]	(d)
[25–35 °C]	[5–25 mm]	[0.2–0.4]	$[5.6\text{--}1.82.10^2]$	[0.021–0.036]	[0.2807–0.4472]	(e)
		[0.4–0.8]	$[5.8\text{--}1.82.10^2]$	[0.0093–0.022]	[0.2807–0.4472]	(f)
	[25–50 mm]	[0.2–0.4]	$[0\text{--}1.82.10^2]$	[0.01–0.42]	[0.2807–0.4472]	(g)
		[0.4–0.8]	$[0\text{--}1.71.10^2]$	[0.0093–0.022]	[0.2807–0.4472]	(h)

This involves evaluating the model's temperature, rainfall, and NDVI-dependent parameters across a defined range of temperatures from 15 °C to 35 °C, rainfall values ranging from 5 mm to 50 mm, and NDVI values ranging from 0.2 to 0.8. We include these ranges in our simulations because a temperature of 16 °C is recognized as the minimum threshold necessary for malaria parasites to mature within mosquitoes [33], and 35 °C represents the maximum temperature for the region under consideration (Burundi, in this case). Subsequently, the range of rainfall is set from 5 mm to 50 mm, as rainfall exceeding 50 mm is considered as the maximum limit for flushing out mosquitoes [13]. Regarding NDVI, we consider both the minimum and maximum NDVI values for Burundi to include the full spectrum of vegetation dynamics within the region. We use the infected class I_H as the response function.

Secondly, we performed the sensitivity analysis when the temperature(T), Rainfall(R) and NDVI(η) are considered as parameters in the ranges of [15–35 °C], [5–50 mm] and [0.2–0.8] respectively. We also used the infected human class as the output. It's important to note that the PRCC values was calculated at five distinct time points: 100, 200, 300, 400, and 500. This analysis over time helps us understand how the model parameters' sensitivity change over the course of the time, giving us a better grasp of the system's dynamic behavior.

Using the population of infectious humans I_H as the response function, parameters β_1 , γ_h , $a(T)$, and $\mu_v(T, \eta)$ are the top common PRCC ranked parameters for all ranges of the

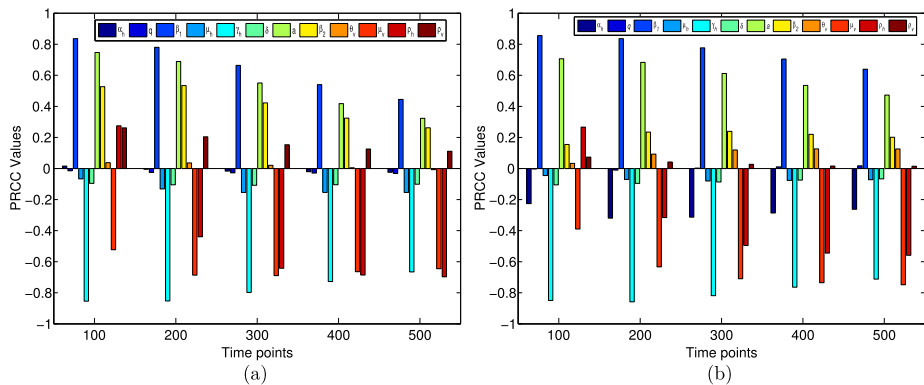


Fig. 4 PRCCs for I_H for different ranges of climate-dependent parameters in the scenarios (a) and (b) of Table 4

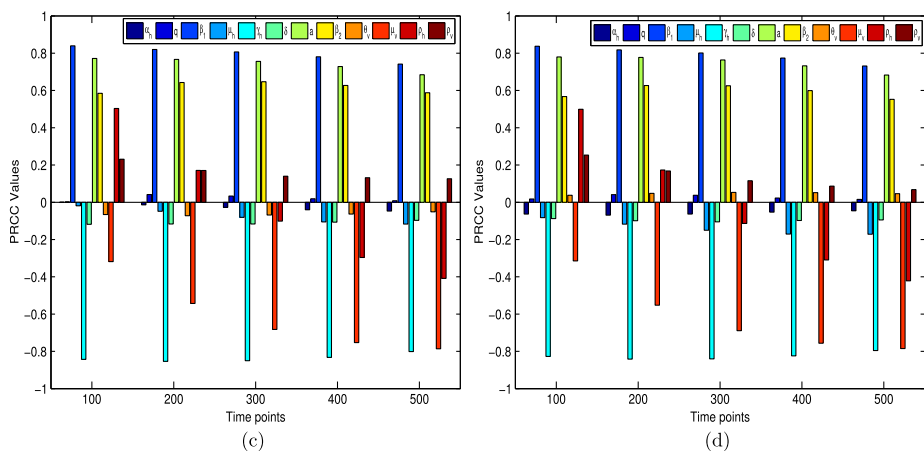


Fig. 5 PRCCs for I_H for different ranges of climate-dependent parameters in the scenarios (c) and (d) of Table 4

climate dependent parameters as established in Table 4. The PRCC values for these parameters may fluctuate a little bit indicating temporal variability in the sensitivity of the model to these parameters. β_1 and $a(T)$ are positively correlated to I_H and thus increase the burden of malaria infection in the human population whereas γ_h and $\mu_v(T, \eta)$ are negatively correlated to I_H and decrease the burden of malaria infection in the human population (Figs. 4 to 7). For scenarios (a), (b), (c), and (d), we observe a slight reduction in β_1 , γ_h , and $a(T)$ over time while there is a significant rise in $\mu_v(T, \eta)$ over time. However, in scenarios (e) and (h), β_1 decreases until it is nearly zero. This coincides to the range of high temperature and high NDVI.

Parameters that exhibit relationships with temperature, rainfall and NDVI over different time periods, sensitive to the model are specifically the mosquito biting rate $a(T)$ and the mosquito mortality rate $\mu_v(T, \eta)$. An increase in the mosquito biting rate $a(T)$ leads to a higher rate of malaria infection in the human population. Conversely, an increase in the mosquito mortality rate $\mu_v(T, \eta)$ has a negative effect on the infection rate, meaning that

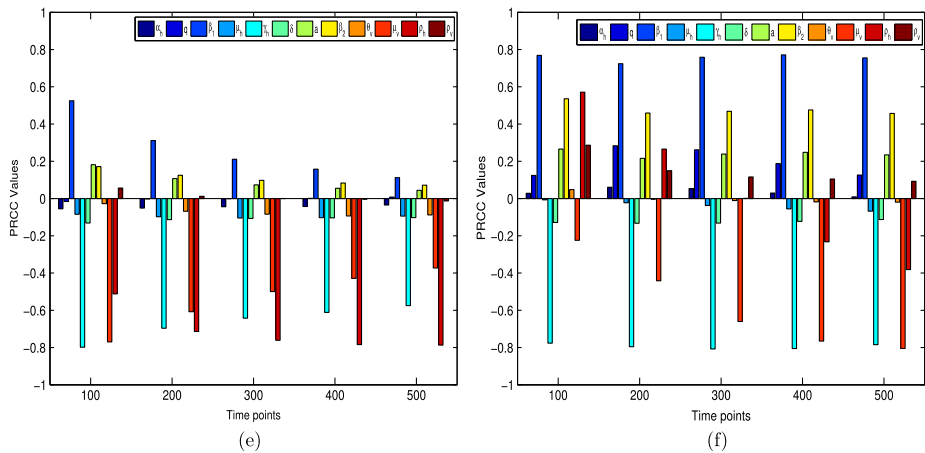


Fig. 6 PRCCs for I_H for different ranges of climate-dependent parameters in the scenarios (e) and (f) of Table 4

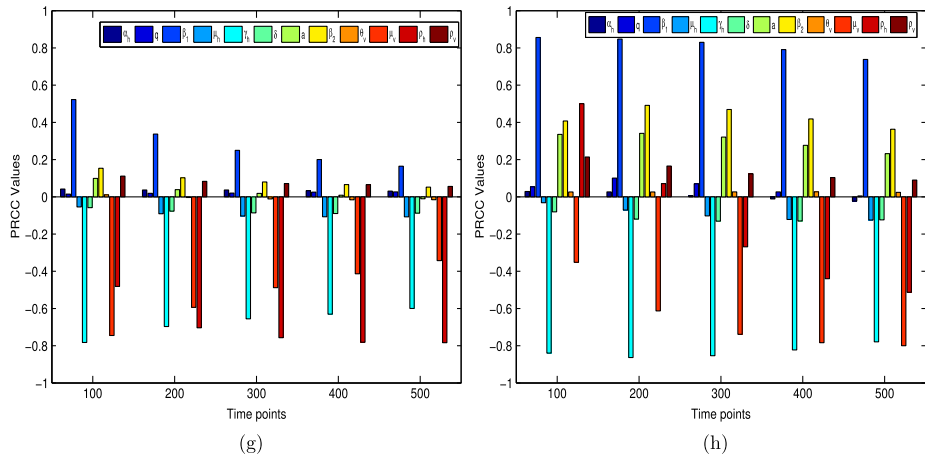
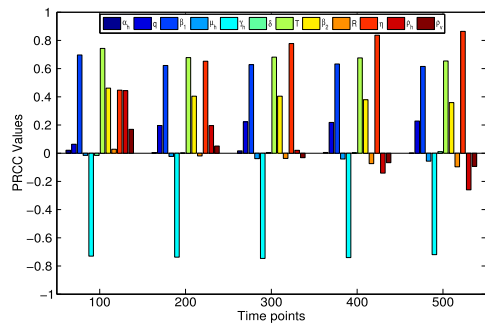


Fig. 7 PRCCs for I_H for different ranges of climate-dependent parameters in the scenarios (g) and (h) of Table 4

a higher μ_v reduces the infection burden. Additionally, we observe that μ_v increases over time, suggesting that the effects of temperature and NDVI on mosquito mortality become more pronounced after several months. This variability underscores the complex nature of malaria's reaction to environmental changes. A comprehensive understanding of these temporal dynamics is pivotal for precise disease prognosis and the development of effective intervention approaches. Consequently, integrating assessments of environmental variability into malaria research is imperative for enhancing our comprehension and management of the disease.

Figure 8 illustrates the PRCC analysis for malaria disease, considering variations in climate variables such as temperature, rainfall, and NDVI. The analysis spans the entire dataset and considers variations in these climate factors. The temperature range investigated lies between 15 °C and 35 °C, the rainfall range considered is 5 mm to 50 mm and the NDVI

Fig. 8 PRCCs for I_H when temperature, rainfall, and NDVI are considered as parameters varying within the system (21)



range explored spans from 0.2 to 0.8. Two key parameters in malaria transmission dynamics, denoted as β_1 and γ_h , exhibit high sensitivity to the variable I_H , with β_1 being positively correlated to I_H and γ_h being negatively correlated to I_H . An increase in β_1 will lead to an increase in I_H , whereas an increase in γ_h will lead to a decrease in I_H .

Temperature and NDVI are positively correlated with I_H . This means that as temperature and NDVI increase, the number of infectious I_H increases. From the analysis over different time points, we observe interesting trends:

- Initially, temperature has a significant impact on I_H . However, as time progresses, its sensitivity decreases slightly. This suggests that other factors might become more influential.
- NDVI's impact on I_H increases over time. This implies that vegetation health (as indicated by NDVI) becomes more relevant in understanding malaria dynamics.
- The sensitivity of rainfall to the output function may not have an impact on the infectious human class for Burundi.

3.3 Simulation Results

3.3.1 Temperature, Rainfall and NDVI Effects on the Model

Figure 9 presents the baseline graphs derived from the mathematical model (21). The simulations are conducted under specific conditions where climate parameters remain fixed. Specifically, the temperature (T) is set at 25 °C, rainfall at 30 mm, and the Normalized Difference Vegetation Index (η) at 0.4. These simulations were carried out over a period of 100 months.

We then illustrate the effects of temperature, rainfall and NDVI on the malaria transmission dynamics. This approach will provide valuable insights into the factors driving malaria propagation. By systematically examining each factor while keeping the others constant, we can gain a better understanding of their individual contributions to malaria transmission. We utilized climate parameter values within the ranges that we used in Sect. 3.2.

To begin, we study the impact of temperature on malaria transmission while keeping rainfall and NDVI constant. Next, we investigated the influence of rainfall on malaria propagation while maintaining constant temperature and NDVI. Afterward, we explored the effects of NDVI on malaria transmission while keeping temperature and rainfall constants (See Figs. 10 to 12)

Figure 10 reveal the trends concerning temperature variations and their impact on the dynamics of malaria transmission within the infected human and infected mosquito individuals. Temperatures below 16 °C are regarded as insufficient for the development of malaria parasite [52]. 16 °C is considered as the minimum temperature for parasite development

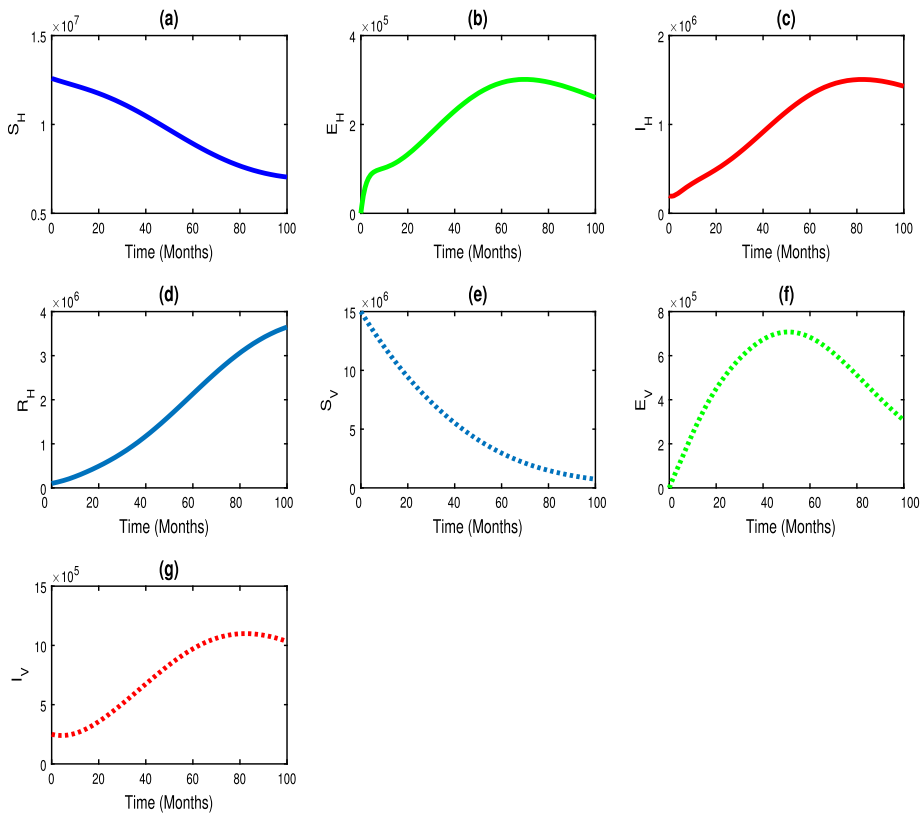


Fig. 9 Baseline graphs shows the dynamics of various state variables within model (21) with $R_0 > 1$ over a period of 100 months

of *Plasmodium falciparum* and *Plasmodium vivax*, limiting the spread of malaria in cooler regions [53].

As shown in Fig. 10, the number of infected humans (I_H) peaks and then gradually decreases at a temperature of $T = 25^\circ\text{C}$ (consistent with the findings in [14, 32]), suggesting that this temperature represents an optimal condition for malaria transmission dynamics. The same figure also indicates that $T = 25^\circ\text{C}$ is optimal for the growth and activity of the infected mosquito population.

This relationship highlights that the significant increase in human infections is directly linked to the growth of mosquito populations at $T = 25^\circ\text{C}$. When environmental conditions are favorable for mosquitoes, their breeding, survival, and feeding rates are enhanced, resulting in higher malaria transmission to the human population.

Figure 11 demonstrates how rainfall affects model (21). It appears that variations in populations due to temperature changes are more significant compared to variations caused by rainfall. The insensitivity of malaria transmission dynamics to varying rainfall levels within the specified range in Burundi suggests that once rainfall exceeds a certain threshold, it does not significantly impact the dynamical transmission of malaria in Burundi (Fig. 11). This is in line with what we found in Sect. 3.2.

In Fig. 12, we were examining the impact of NDVI. We found that NDVI has a greater influence on the model compared to rainfall. This can be explained by the fact that Vegetation

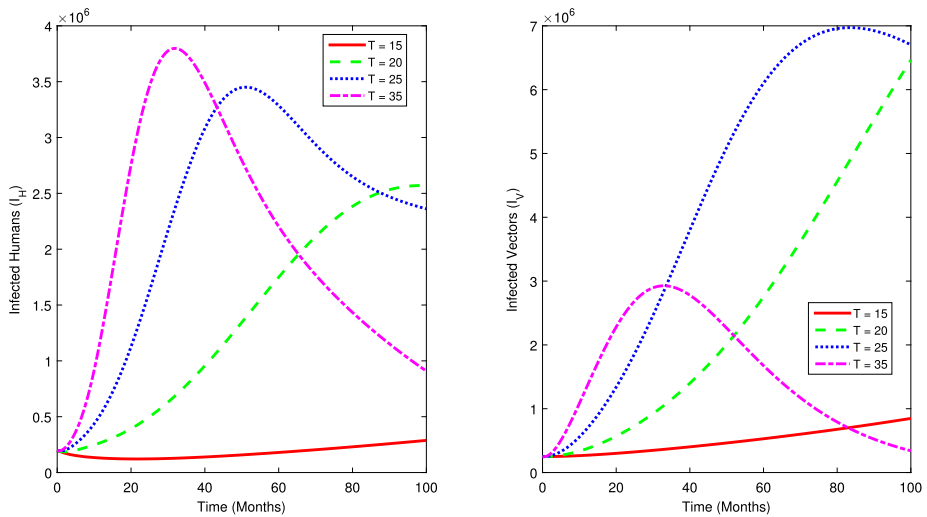


Fig. 10 Graphical illustration of model (21) for different temperatures when NDVI and rainfall are constant

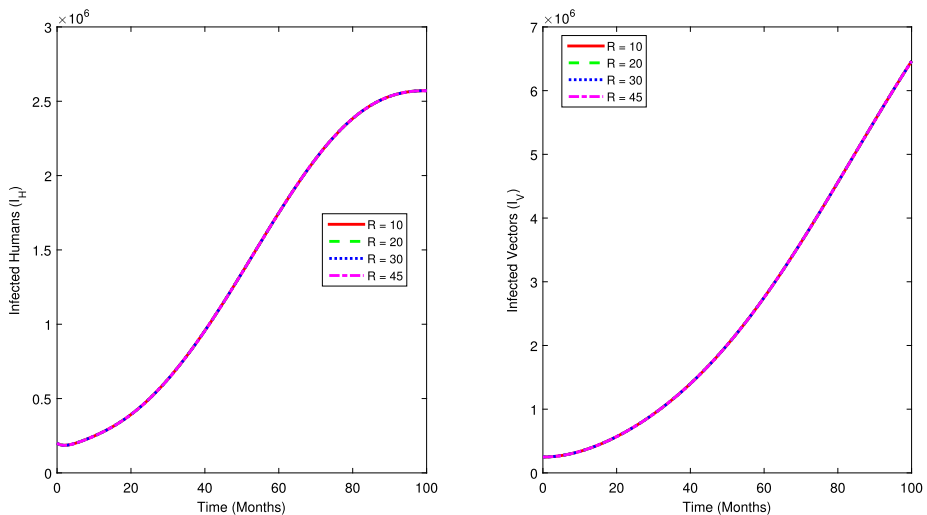


Fig. 11 Graphical illustration of model (21) for different rainfall when temperature and NDVI are constant

influences not only the availability of breeding sites but also the microclimate (humidity and temperature), which can affect mosquito survival and reproduction. In regions with dense vegetation, even if rainfall is inconsistent, mosquitoes can still find suitable conditions for reproduction [54]. This might make NDVI a more consistent and powerful predictor than rainfall in the model. When NDVI increases from 0.2 to 0.8, we observe an increase in infectious individuals, both humans and mosquitoes respectively. We observed a high number in the infected individuals in the range of [0.4–0.6]. The level of vegetation (NDVI) in an area seems to have a stronger impact on the model's predictions than the amount of rainfall.

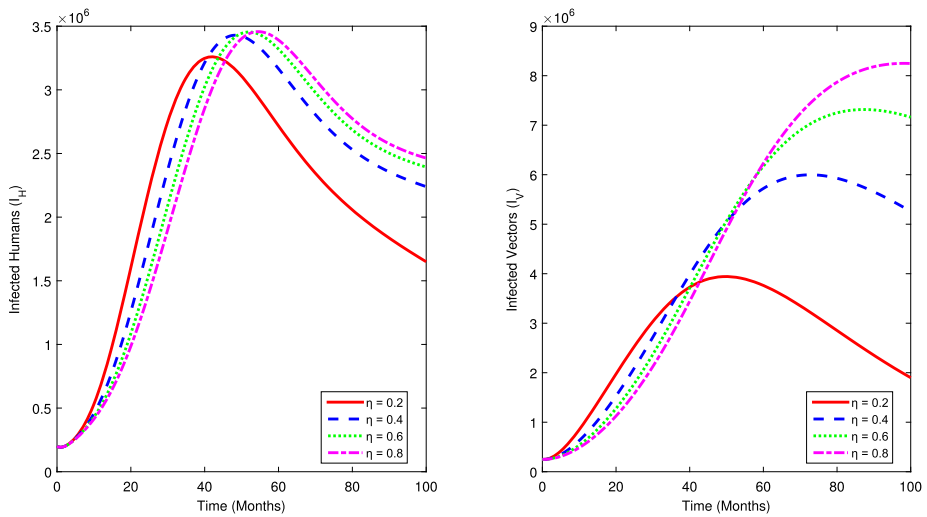


Fig. 12 Graphical illustration of model (21) for different values of NDVI when temperature and rainfall are constant

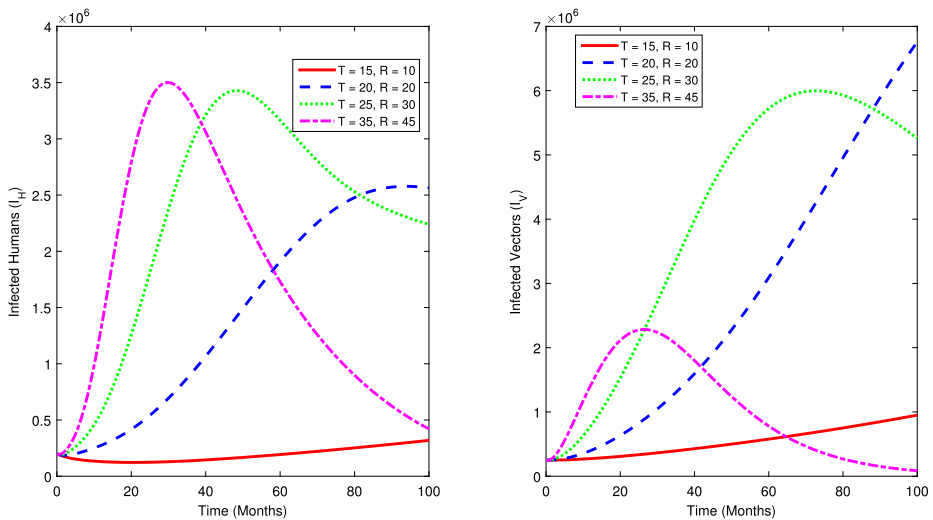


Fig. 13 Graphical illustration of model (21) for different values of temperature and rainfall when NDVI is constant

After showing the variation of a single parameter while maintaining the others constants, we proceeded to simultaneously modify two parameters while keeping one constant. From the simulations, we made interesting observations. When combining temperature with rainfall or NDVI (see Fig. 13 and Fig. 14), we observe significant changes in the transmission dynamics of malaria. But when rainfall was combined with NDVI (Fig. 15), the alterations in transmission dynamics of malaria were relatively small. With low temperatures combined to low rainfall or low NDVI, the infected populations remain nearly constant. This can be attributed to the fact that the growth rate of parasites in mosquitoes is considered to be mini-

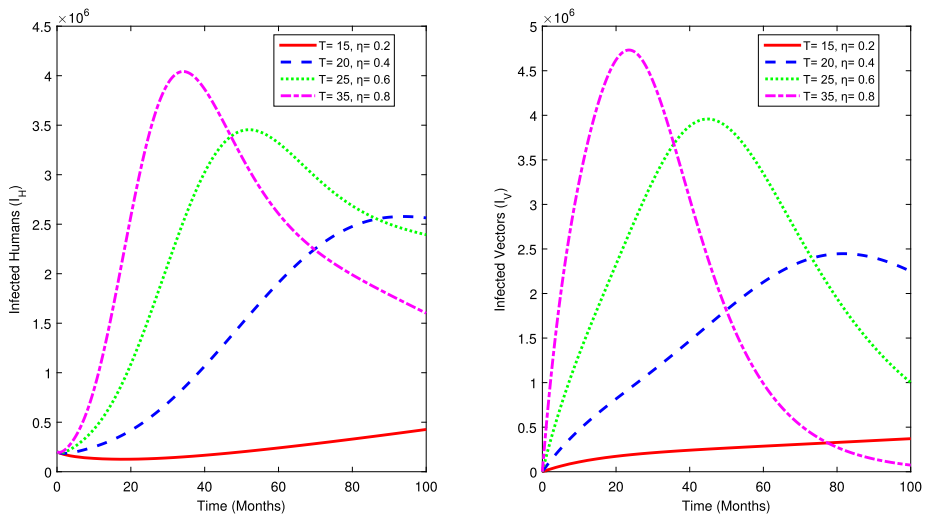


Fig. 14 Graphical illustration of model (21) for different values of temperature and NDVI when rainfall is constant

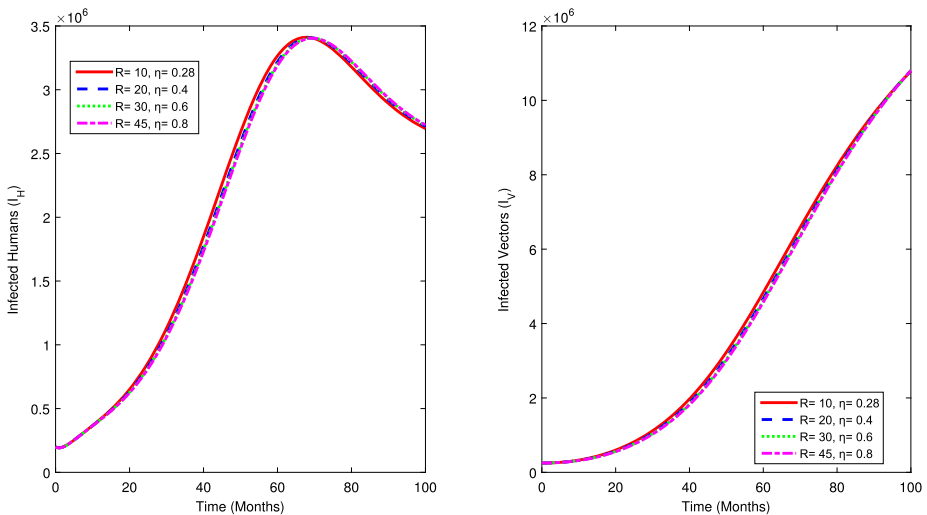


Fig. 15 Graphical illustration of model (21) for different values of rainfall and NDVI when temperature is constant

mal when the temperature is below 16 °C. As the temperature rises, both the infected human and mosquito populations begin to decrease. The number of cases in the infection classes reached their peak at a temperature value range of [20–25 °C], in conjunction with a mean monthly range of [20–30 mm] and NDVI range of [0.4–0.6]. When we combine rainfall and NDVI, we observe only minor changes in the populations.

In the scenario where temperature, rainfall, and NDVI are all fluctuating, we observed that where all three variables are low, the infection classes remain relatively constant, but they begin to increase as the values of temperature, rainfall, and NDVI increase. This in-

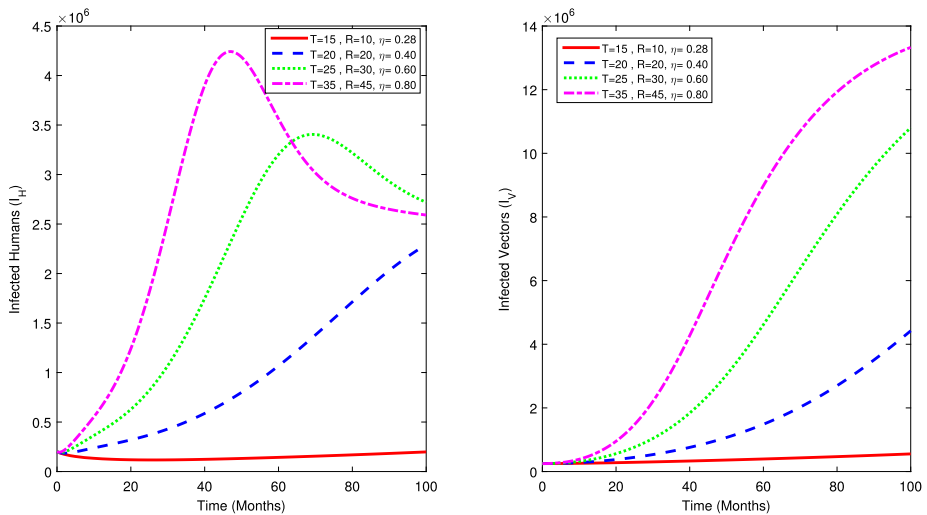


Fig. 16 Graphical illustration of model (21) for different values of Temperature, Rainfall and NDVI

crease becomes more pronounced, ultimately reaching a maximum at a temperature of 35, rainfall of 45 mm, and an NDVI value of 0.8 (Fig. 16).

3.4 Forecasting Temporal Trends in the Reproduction Number

Mathematical models are valuable tools for understanding and predicting disease risks, including malaria. A key parameter in these models is the basic reproduction number (R_0), which quantifies the potential risk for disease transmission. The reproduction number (R_0) is a critical factor in understanding disease spread and evaluating the effectiveness of control measures [55]. It can provide a numerical basis for predicting how climate fluctuations can impact disease transmission. In the context of malaria, R_0 is influenced by climate and environmental factors. By calculating R_0 under various climate conditions, researchers can qualitatively assess future malaria risk and inform control strategies. We use projected temperature, rainfall, and NDVI data for Burundi to predict malaria infection.

To illustrate the evolution of climate variables over time, we present future outcomes in comparison to the reference period (2010–2020). By carefully analyzing the differences between future climate patterns and those recorded during the reference period, we can determine the extent and characteristics of long-term climate fluctuations and their impacts. Here we used three future climate periods, period 2021 to 2050, period of 2031 to 2060 and period of 2041 to 2070. We choose these periods following the recommendations of the World Meteorological Organization (WMO) [56].

We observed that when using R_0 as a risk index for malaria transmission dynamics under the influence of temperature, rainfall, and NDVI, malaria severity in Burundi exhibits distinct peaks in January, April, and December. After January, there is a slight decrease before a rise leading to the peak in April, followed by another increase in August, culminating in the peak in December. In [57], statistical descriptive analysis of malaria cases showed that malaria infections peak in January, decrease in February, rise from March to June, drop sharply in August, and then increase again from September to December. We observed that both results highlight the significant seasonal variations in malaria transmission, emphasizing some similar periods of increased and decreased malaria cases throughout the year.

Fig. 17 Basic reproduction number based on temperature, rainfall and NDVI climate data

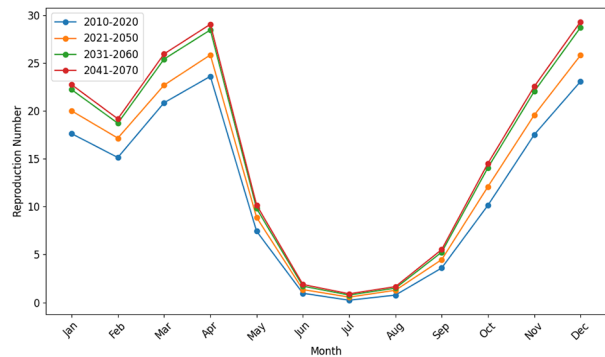


Figure 17 shows an increase in R_0 over time from 2010 to 2070, indicating that the disease is becoming more transmissible over time. This is concerning because a higher reproduction number means that more people are likely to be infected, leading to a larger outbreak or epidemic if effective control measures are not put in place.

3.5 Discussion of Results

We presented a mathematical model that explores the influence of temperature, rainfall, and NDVI on malaria transmission. The model is formulated and is thoroughly examined. We employed mathematical techniques to determine both the local stability of the disease-free and endemic equilibrium and the global stability of the disease-free equilibrium. Our analysis produced mathematical expressions that describe the conditions under which the disease will either continue to spread or be brought under control.

The analysis conducted by PRCCs suggested that the mosquito-human contact rate has a positive effect on malaria incidence, implying that any factor reducing this contact rate could help mitigate the spread of malaria. Conversely, the mortality rate of mosquitoes has a negative correlation with the malaria incidence, indicating that anything increasing this rate could lower malaria cases. In Burundi, distribution of long-lasting insecticidal nets (LLINs) started in the year 2003 [58] but LLINs are reported to provide effective protection against malaria for only one year. This limited duration of efficacy is crucial for understanding the frequency and timing of distribution campaigns needed to maintain high levels of protection within the population [59]. Interventions such as intermittent preventive treatment in pregnancy and bed nets distributed during antenatal care visits and delivery started in the year 2006 helped in the reduction of cases in malaria transmission for pregnant women [58]. So, other interventions such as indoor and outdoor residual spraying made regularly, have the potential to reduce malaria transmission in the population of Burundi. It also showed that temperature and NDVI appears to increase malaria incidence compared to rainfall in Burundi.

Numerical simulations were carried out on the model to investigate the influence of temperature, rainfall and NDVI on malaria transmission. We used monthly scale as changes in vegetation cannot be seen daily or weekly. Mean monthly temperature ranges from 15 °C to 35 °C were used, revealing that malaria transmission is most effective in the range of [20–25 °C], aligning with previous research [32, 60], when rainfall and NDVI are constant. Mean monthly rainfall values below 50 mm were examined, considering that mosquito breeding sites are flushed out with rainfall exceeding 50 mm [13]. Despite changes in rainfall levels, only minor differences in populations are observed. The simulation results showed that rainfall has little impact on malaria transmission dynamics in Burundi. Monthly NDVI values

from 0.2 to 0.8 were considered in simulations whereas temperature and rainfall are constants. The results showed that NDVI range of [0.4-0.6] is the most favorable to malaria transmission. A study in Uganda showed that the average NDVI of 0.72 significantly increased the cumulative incidence rate ratio (IRR) of malaria [61]. Our study seems to suggest that the addition of temperature and rainfall in the model would require slightly less NDVI levels to increase the transmission of malaria. The increase of all the three variables temperature, rainfall and NDVI led to an increase in infection cases. This could potentially indicate a correlation between environmental conditions and the spread of infections.

In numerous studies, the optimal conducive temperature range for malaria transmission has been identified as 20-32 °C [32, 60, 62]. However, our model indicates that when additional variables such as NDVI and rainfall are incorporated, the effective temperature range may shift, where even at higher temperature than 32 °C, the level of infection may continue to increase. This was the observation when the model was calibrated with the information from Burundi. Mathematical models should integrate the effects of multiple variables, including NDVI and rainfall, to provide a more comprehensive understanding of malaria transmission dynamics.

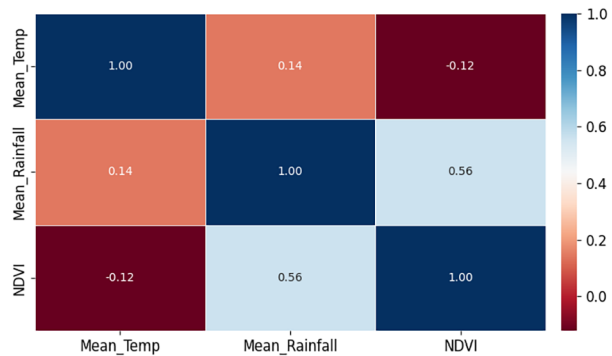
We also used the reproduction number (R_0) as a fundamental measure to assess how climate variations affect the dynamics of malaria transmission. Specifically, we examined the impact of temperature, rainfall, and NDVI on R_0 and subsequently on the transmission of malaria dynamics. Through the analysis of climate projections for the period between 2021 and 2070, our aim was to predict potential changes in the severity of malaria using the values of R_0 . Our results indicated a significant increase in R_0 over time, suggesting a rise in the ability of the disease to spread in future. Interventions such as spraying and utilizing treated bed nets, which aid in reducing mosquito populations, should be promoted, particularly in the time when reproduction number is high and when climatic conditions are conducive for mosquito breeding. These results underscore the threat of larger outbreaks or epidemics in the future, emphasizing the urgent necessity for effective strategies to mitigate malaria transmission in response to evolving climate conditions.

We have presented a basic model capturing the scenarios for Burundi and this is one of the few studies on malaria transmission dynamics influenced by the three climatic factors. The study is not all encompassing and there is a room for improvement on future research which should incorporate seasonality into the model to get a more comprehensive evaluation as certain climate factors exhibiting stronger connections with malaria transmission over longer time frames. This approach will enable capturing the impacts that unfold over extended periods, rather than being confined to short-term fluctuations within a single month. Through the adoption of this more precise methodology, we can discern clearer patterns and obtain a deeper understanding of the complex relationship between climate variables and the dynamics of malaria. By also using provincial-level data instead of national-level data, we can pinpoint regions where malaria is likely to increase in the future and concentrate control efforts accordingly.

Appendix A: Correlation Analysis

Using the Spearman method, we aim to discern any significant associations between temperature, rainfall, and NDVI, which can provide valuable insights into potential climatic relationships with vegetation health and density. The analysis indicates that while there are discernible associations between these variables, they tend to be relatively weak. Mean

Fig. 18 Correlation matrix between mean monthly temperature, mean monthly rainfall and NDVI



monthly temperature exhibits a slight positive correlation with mean monthly rainfall, implying that areas experiencing higher temperatures may also see slightly increased rainfall, though this connection is not strong. Conversely, the mean monthly temperature displays a modest negative correlation with NDVI, indicating that regions with higher temperatures may have slightly lower vegetation density, though again, this association is not substantial. The most notable relationship emerges between mean monthly rainfall and NDVI, with a positive correlation indicating that areas with greater rainfall tend to exhibit higher vegetation density. This result aligns with expectations, as increased rainfall typically supports more lush vegetation growth.

Appendix B: Forecasting Data: Normalized Vegetation Index

The process of projecting with a SARIMA model involves utilizing historical NDVI values to forecast future values, capturing patterns such as seasonality and autocorrelation for predictions. To do so we used NDVI data from 2010 to 2022 Fig. 2c

The SARIMA model was then employed to analyze and forecast seasonal and temporal patterns inherent in the NDVI data. This robust technique accounts for both seasonal and non-seasonal variations, crucial for accurately capturing underlying trends. Model fitting involved selecting appropriate parameters including autoregressive (AR), differencing (I), moving average (MA), and seasonal components based on NDVI time series characteristics. The SARIMA model extends ARIMA by incorporating seasonal components like seasonal autoregressive (SAR) and seasonal moving average (SMA) components. The autoregressive component captures the linear relationship between current and previous observations, while differencing accounts for non-stationarity. The model's parameters are estimated based on observed data to make future forecasts while considering seasonal fluctuations. Rigorous validation procedures, including statistical metrics and graphical analyses, evaluated the model's effectiveness in capturing NDVI seasonality and temporal dynamics, providing reliable forecasts crucial for environmental and ecological applications.

B.1 Model Identification

We found that the Mean Absolute Error (MAE) of approximately 0.01675 and the R-squared (rsquare) value of approximately 0.9344 provide insightful metrics for assessing the performance of the SARIMA model on the validated set spanning from January 2020 to December 2022. These results collectively indicate that the SARIMA model performs well in forecasting NDVI values.

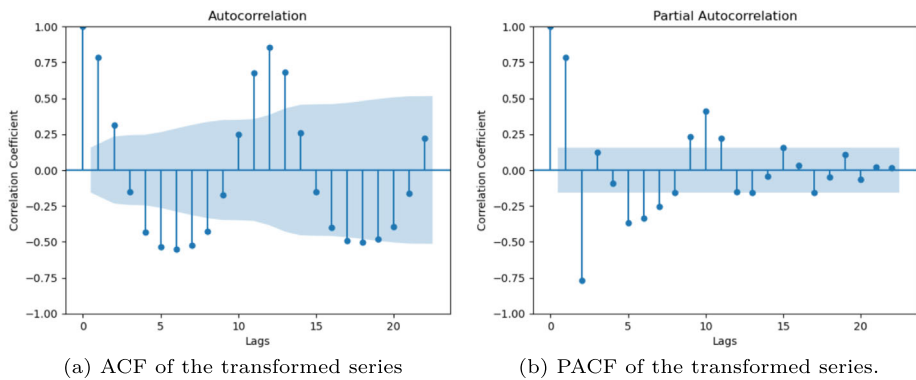
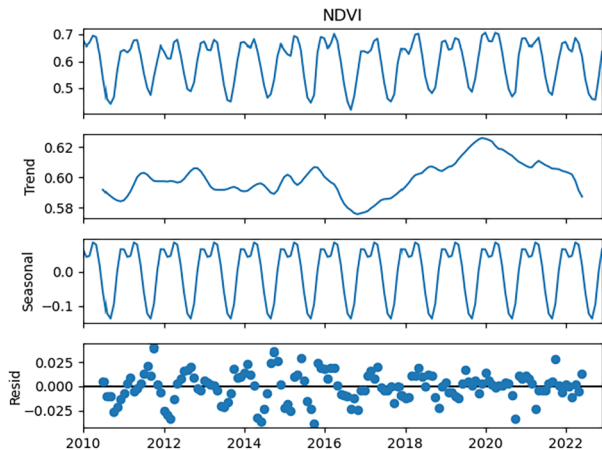


Fig. 19 ACF and PACF of the transformed series

Fig. 20 Decomposition analysis of the NDVI



B.2 Validation of the Model

Validation of the SARIMA model's performance was conducted using a separate dataset from January 2020 to December 2022, assessing its ability to generalize data and forecasting accuracy.

The SARIMA results provide a comprehensive insight into the NDVI time series dynamics. Notably, the parameter estimates reveal significant relationships between the current NDVI values and their lagged counterparts, as well as the residual errors from previous forecasts. For instance, the coefficient of $ar.L1$ (-0.6165) suggests a negative correlation between the current NDVI value and its lagged value, while the coefficient of $ar.L2$ (0.7379) indicates a positive correlation with the value two time periods ago. After estimating the parameters of the SARIMA (4,1,3)(2,0,3,12) model, we evaluated their adequacy by examining the residuals.

Measures of model fit such as the log likelihood (-614.852), AIC (-578.723), BIC, and HQIC indicate a well-fitted SARIMA model to the NDVI data, with lower values suggesting superior performance. Diagnostic tests for autocorrelation, normality, and heteroskedasticity validate the model's adequacy in capturing temporal dependencies and distributional assumptions. With the high p-values associated with the statistics, we fail to reject the null

Fig. 21 The graphical representation of the validation process of the seasonal autoregressive integrated moving average (SARIMA) model

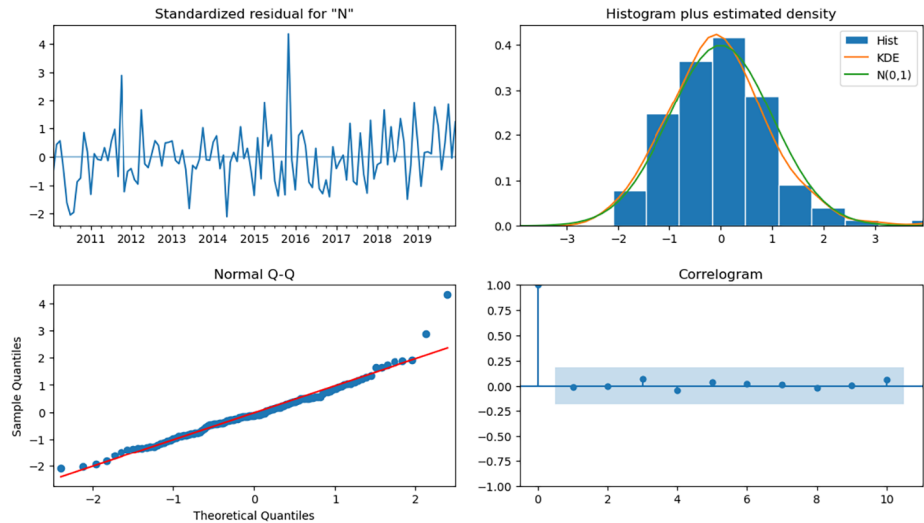
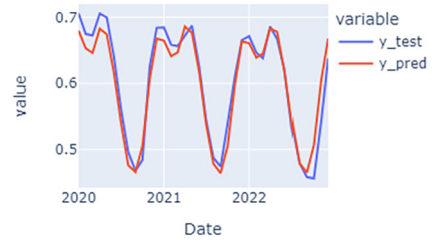
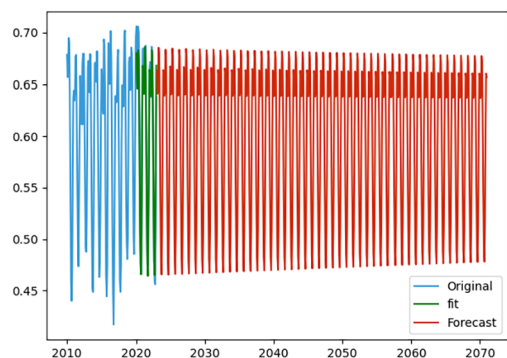


Fig. 22 Graphical diagnostics for assessing the SARIMA (4,1,3)(2,0,3,12) model fit

Fig. 23 NDVI forecasted



hypothesis of independence in this residual series. Thus, we conclude that the SARIMA model (4,1,3)(2,0,3,12) adequately fits the data. Figure 20 shows the forecasted NDVI data (in red) from January 2023 to December 2070

Following the same procedure, we forecast rainfall and Temperature.

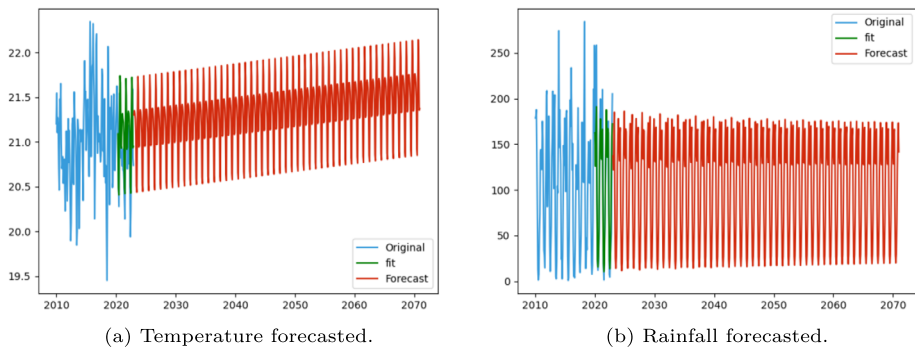


Fig. 24 Forecasted temperature and rainfall

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Data Availability The NDVI data were extracted using Google Earth Engine [63]. Both the malaria data (from INMCP) and the temperature and rainfall data (from IGEBU) come from inhouse data for the wide project on Malaria epidemiology in Burundi led by the Doctoral school at the University of Burundi. The data are available for researchers but not for public.

Declarations

Competing Interests The authors declare that there is no conflict of interest regarding the publication of this paper.

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References

1. Djihinto, O.Y., Medjigbodo, A.A., Gangbadja, A.R.A., Saizonou, H.M., Lagnika, H.O., Nanmede, D., Djossou, L., Bohouton, R., Sovegnon, P.M., Fanou, M.-J., et al.: Malaria-transmitting vectors microbiota: overview and interactions with anopheles mosquito biology. *Front. Microbiol.* **13**, 891573 (2022)
2. Zongo, P.: Modélisation mathématique de la dynamique de la transmission du paludisme. PhD thesis, Université de Ouagadougou (2009)
3. Sato, S.: Plasmodium—a brief introduction to the parasites causing human malaria and their basic biology. *J. Physiol. Anthropol.* **40**(1), 1–13 (2021)
4. World Health Organization: World Malaria Report 2022. World Health Organization, Geneva, Switzerland (2022). <https://www.who.int/publications/i/item/9789240064898>

5. Venkatesan, P.: The 2023 WHO World malaria report. *The Lancet Microbe* (2024)
6. Lok, P., Dijk, S.: Malaria outbreak in Burundi reaches epidemic levels with 5.7 million infected this year. *Br. Med. J.* (2019)
7. Mohanan, P., Islam, Z., Hasan, M.M., Adedeji, O.J., Santos Costa, A.C., Aborode, A.T., Ahmad, S., Essar, M.Y.: Malaria and covid-19: a double battle for burundi. *Afr. J. Emerg. Med.* **12**(1), 27–29 (2022)
8. World Vision International: Eight facts about Burundi's malaria epidemic (2017). Retrieved from <https://www.wvi.org/article/8-facts-about-burundis-malaria-epidemic>
9. Moise, I.K., Roy, S.S., Nkengurutshe, D., Ndikubagenzi, J.: Seasonal and geographic variation of pediatric malaria in Burundi: 2011 to 2012. *Int. J. Environ. Res. Public Health* **13**(4), 425 (2016)
10. LaPointe, D.A., Atkinson, C.T., Samuel, M.D.: Ecology and conservation biology of avian malaria. *Ann. N.Y. Acad. Sci.* **1249**(1), 211–226 (2012)
11. Tolle, M.A.: Mosquito-borne diseases. *Curr. Prob. Pediatr. Adolesc. Health Care* **39**(4), 97–140 (2009)
12. Martens, W.J.M., Jetten, T.H., Rotmans, J., Niessen, L.W.: Climate change and vector-borne diseases: a global modelling perspective. *Glob. Environ. Change* **5**(3), 195–209 (1995)
13. Parham, P.E., Michael, E.: Modeling the effects of weather and climate change on malaria transmission. *Environ. Health Perspect.* **118**(5), 620–626 (2010)
14. Yiga, V., Nampala, H., Tumwiine, J.: Analysis of the Model on the Effect of Seasonal Factors on Malaria Transmission Dynamics. *J. Appl. Math.* (2020). Hindawi
15. Githeko, A.K., Lindsay, S.W., Confalonieri, U.E., Patz, J.A.: Climate change and vector-borne diseases: a regional analysis. *Bull. World Health Organ.* **78**(9), 1136–1147 (2000). SciELO Public Health
16. Gubler, D.J., Reiter, P., Ebi, K.L., Yap, W., Nasci, R., Patz, J.A.: Climate variability and change in the United States: potential impacts on vector-and rodent-borne diseases. *Environ. Health Perspect.* **109**(Suppl 2), 223–233 (2001)
17. Mabaso, M.L.H., Craig, M., Ross, A., Smith, T.: Environmental predictors of the seasonality of malaria transmission in Africa: the challenge. *Am. J. Trop. Med. Hyg.* **76**(1), 33–38 (2007). American Society of Tropical Medicine and Hygiene
18. Ezeakacha, N.F., Yee, D.A.: The role of temperature in affecting carry-over effects and larval competition in the globally invasive mosquito *Aedes albopictus*. *Parasites Vectors* **12**, 1–11 (2019). Springer
19. Couper, L.I., Farner, J.E., Caldwell, J.M., Childs, M.L., Harris, M.J., Kirk, D.G., Nova, N., Shocket, M., Skinner, E.B., Uricchio, L.H., Exposito-Alonso, M., Mordecai, E.A.: How will mosquitoes adapt to climate warming? *eLife* **10**, e69630 (2021)
20. Eisele, T.P., Keating, J., Swalm, C., Mbogo, C.M., Githeko, A.K., Regens, J.L., Githure, J.I., Andrews, L., Beier, J.C.: Linking field-based ecological data with remotely sensed data using a geographic information system in two malaria endemic urban areas of Kenya. *Malar. J.* **2**(1), 1–17 (2003). BioMed Central
21. Wayant, N.M., Maldonado, D., de Arias, A.R., Cousiño, B., Goodin, D.G.: Correlation between normalized difference vegetation index and malaria in a subtropical rain forest undergoing rapid anthropogenic alteration. *Geosp. Health* **4**(2), 179–190 (2010)
22. Gaudart, J., Touré, O., Dessay, N., Dicko, A.L., Ranque, S., Forest, L., Demongeot, J., Doumbo, O.K.: Modelling malaria incidence with environmental dependency in a locality of Sudanese savannah area, Mali. *Malar. J.* **8**, 1–12 (2009). Springer
23. McKenzie, F.E.: Why model malaria? *Parasitol. Today* **16**(12), 511–516 (2000). Elsevier
24. Abboubakar, H., Buonomo, B., Chitnis, N.: Modelling the effects of malaria infection on mosquito biting behaviour and attractiveness of humans. *Ric. Mat.* **65**(1), 329–346 (2016). Springer
25. Kouchère, A., Abboubakar, H., Damakoa, I.: Analysis of a mosquito life cycle model. *ARIMA* **34** (2021). Episciences.org
26. Koella, J.C.: On the use of mathematical models of malaria transmission. *Acta Trop.* **49**(1), 1–25 (1991). Elsevier
27. Ross, R.: *The Prevention of Malaria*. John Murray, London (1911)
28. Anderson, R.M., May, R.M.: *Infectious Diseases of Humans: Dynamics and Control*. Oxford University Press, Oxford (1992)
29. Koella, J.C., Boete, C.: A model for the coevolution of immunity and immune evasion in vector-borne diseases with implications for the epidemiology of malaria. *Am. Nat.* **161**(5), 698–707 (2003). The University of Chicago Press
30. Kribs-Zaleta, C.M., Martcheva, M.: Vaccination strategies and backward bifurcation in an age-since-infection structured model. *Math. Biosci.* **177**, 317–332 (2002). Elsevier
31. Macdonald, G.: *The Epidemiology and Control of Malaria*. Oxford University Press, London (1957)
32. Abiodun, G.J., Njabo, K.Y., Witbooi, P.J., Adeola, A.M., Fuller, T.L., Okosun, K.O., Makinde, O.S., Botai, J.O.: Exploring the influence of daily climate variables on malaria transmission and abundance of *Anopheles arabiensis* over Nkomazi local municipality, Mpumalanga province, South Africa. *J. Environ. Public Health* (2018). Hindawi

33. Abdelrazec, A., Gumel, A.B.: Mathematical assessment of the role of temperature and rainfall on mosquito population dynamics. *J. Math. Biol.* **74**, 1351–1395 (2017)
34. Dantur Juri, M.J., Estallo, E., Almirón, W., Santana, M., Sartor, P., Lamfri, M., Zaidenberg, M.: Satellite-derived NDVI, LST, and climatic factors driving the distribution and abundance of anopheles mosquitoes in a former malarious area in northwest Argentina. *J. Vector Ecol.* **40**(1), 36–45 (2015)
35. Amadi, J.A., Olago, D.O., Ong'amo, G.O., Oriaso, S.O., Nanyingi, M., Nyamongo, I.K., Estambale, B.B.A.: Sensitivity of vegetation to climate variability and its implications for malaria risk in Baringo, Kenya. *PLoS ONE* **13**(7) (2018)
36. Ferraccioli, F., Riccetti, N., Fasano, A., Mourelatos, S., Kioutsoukis, I., Stilianakis, N.I.: Effects of climatic and environmental factors on mosquito population inferred from West Nile virus surveillance in Greece. *Sci. Rep.* **13**(1), 18803 (2023). Nature Publishing Group UK London
37. Graumans, W., Jacobs, E., Bousema, T., Sinnis, P.: When is a plasmodium-infected mosquito an infectious mosquito? *Trends Parasitol.* **36**(8), 705–716 (2020)
38. Diekmann, O., Heesterbeek, J.A.P., Metz, J.A.J.: On the definition and the computation of the basic reproduction ratio R_0 in models for infectious diseases in heterogeneous populations. *J. Math. Biol.* **28**(4), 365–382 (1990)
39. van den Driessche, P., Watmough, J.: Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission. *Math. Biosci.* **180**(1–2), 29–48 (2002)
40. Mandal, S., Sarkar, R.R., Sinha, S.: Mathematical models of malaria - a review. *Malar. J.* **10**(1), 1–19 (2011). BioMed Central
41. Chowell, G., Diaz-Duenas, P., Miller, J.C., Alcazar-Velazco, A., Hyman, J.M., Fenimore, P.W., Castillo-Chavez, C.: Estimation of the reproduction number of Dengue fever from spatial epidemic data. *Math. Biosci.* **208**(2), 571–589 (2007)
42. van den Driessche, P.: Reproduction numbers of infectious disease models. *Infect. Dis. Model.* **2**(3), 288–303 (2017). <https://doi.org/10.1016/j.idm.2017.06.002>
43. Bavafa-Toosi, Y.: Introduction to Linear Control Systems, pp. 265–331. Academic Press, San Diego (2019). <https://doi.org/10.1016/B978-0-12-812748-3.00003-3>
44. Castillo-Chavez, C., Feng, Z., Huang, W.: On the computation of R_0 and its role on global stability. In: *Mathematical Approaches for Emerging and Re-Emerging Infection Diseases: An Introduction*, vol. 125, pp. 31–65 (2002)
45. Juga, M.L., Nyabadza, F., Chirove, F.: An Ebola virus disease model with fear and environmental transmission dynamics. *Infect. Dis. Model.* **6**, 545 (2021)
46. Castillo-Chavez, C., Song, B.: Dynamical models of tuberculosis and their applications. *Math. Biosci. Eng.* **1**(2), 361–404 (2004)
47. Bari, E., Nipa, N.J., Roy, B.: Association of vegetation indices with atmospheric & biological factors using MODIS time series products. *Environ. Chall.* **5**, 100376 (2021)
48. Abraham, B.O., Ledolter, J.: Seasonal autoregressive integrated moving average models. In: *Statistical Methods for Forecasting* (2008)
49. World Population Review: Burundi Population 2024 (Live) (n.d.). Accessed on: 2024-05-17. Available at <https://worldpopulationreview.com/countries/burundi>
50. Blayneh, K., Cao, Y., Kwon, H.D.: Optimal control of vector-borne diseases: treatment and prevention. *Discrete Contin. Dyn. Syst., Ser. B* **11**(3), 587–611 (2009)
51. Gomerio, B.: Latin Hypercube Sampling and Partial Rank Correlation Coefficient Analysis Applied to an Optimal Control Problem. Master's Thesis, University of Tennessee (2012). https://trace.tennessee.edu/utk_gradthes/1278
52. Waite, J.L., Suh, E., Lynch, P.A., Thomas, M.B.: Exploring the lower thermal limits for development of the human malaria parasite, plasmodium falciparum. *Biol. Lett.* **15**(6), 20190275 (2019)
53. Patz, J.A., Olson, S.H.: Malaria risk and temperature: influences from global climate change and local land use practices. *Proc. Natl. Acad. Sci. USA* **103**(15), 5635–5636 (2006)
54. Smith, D.C., Schäfer, S.M., Golding, N., Nunn, M.A., White, S.M., Callaghan, A., Purse, B.V.: Vegetation structure drives mosquito community composition in UK's largest managed lowland wetland. *Parasites Vectors* **17**(1), 201 (2024)
55. Greene, J.A., Vargha, D.: Ends of epidemics. In: *COVID-19 and World Order: The Future of Conflict, Competition, and Cooperation*, pp. 23–39 (2020)
56. Fathauer, T.: A glimpse into a changing climate: new 1981–2010 climate normals take effect. *Weatherwise* **64**(6), 34–36 (2011)
57. Mfisismana, L.D., Nibayisabe, E., Badu, K., Niyukuri, D.: Exploring predictive frameworks for malaria in Burundi. *Infect. Dis. Model.* **7**(2), 33–44 (2022)
58. Sinzinkayo, D., Baza, D., Nganguenon, V., Koepfli, C.: The lead-up to epidemic transmission: malaria trends and control interventions in Burundi 2000 to 2019. *Malar. J.* **20**(1), 298 (2021)

59. Van Bortel, W., Mariën, J., Jacobs, B.K.M., Sinzinkayo, D., Sinarinzi, P., Lampaert, E., D'hondt, R., Mafuko, J.-M., De Weggheleire, A., Vogt, F., et al.: Long-lasting insecticidal nets provide protection against malaria for only a single year in Burundi, an African highland setting with marked malaria seasonality. *BMJ Glob. Health* **7**(12), e009674 (2022)
60. Ngarakana-Gwasira, E.T., Bhunu, C.P., Mashonjowa, E.: Assessing the impact of temperature on malaria transmission dynamics. *Afr. Math.* **25**(4), 1095–1112 (2014)
61. Okiring, J., Routledge, I., Epstein, A., Namuganga, J.F., Kamya, E.V., Obeng-Amoako, G.O., Sebuguzi, C.M., Rutazaana, D., Kalyango, J.N., Kamya, M.R., et al.: Associations between environmental covariates and temporal changes in malaria incidence in high transmission settings of Uganda: a distributed lag nonlinear analysis. *BMC Public Health* **21**, 1–11 (2021)
62. Okuneye, K., Gumel, A.B.: Analysis of a temperature-and rainfall-dependent model for malaria transmission dynamics. *Math. Biosci.* **287**, 72–92 (2017)
63. Google Earth Engine: Google Earth Engine (n.d.). Accessed on: 2024-07-19. Available at <https://earthengine.google.com/>

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